

Supplementary Material for “Observational Indistinguishability and Integrity Blind Regions in Hybrid Quantum-Classical Workflows”

Roberto Fernández-Barrios, Iker Pastor-López, Amaia Pikatza-Huerga, and Pablo García Bringas
Faculty of Engineering, University of Deusto, Bilbao, Spain



1 PURPOSE AND CLAIM BOUNDARY

This supplement records the complete formal statements with proofs, the experimental design, the adversary-model reading rule, the statistical units and their dependence structure, the validation checks, the full tables of the pre-registered gates of artifacts 1.1.0, 1.2.0 and 1.3.0 (including the amendment that replaced the calibration rule and the adversarial cluster-preserving gate), the secondary model-impact profile and the reproduction mapping behind the main article. The frozen geometry-aligned gate adds one preregistered geometry-aligned sensitivity analysis relating clean-resample calibration to same-item intervention scoring. It changes no dataset, model family, kernel, feature map, hyperparameter search, threat class, formal theorem or primary structural experiment. It adds no QPU, calibrated-noise, scheduling, provider-security, multi-tenancy, deployed runtime-service or operational Fleet Management evidence. The exact claims remain bounded to data-to-evaluation interventions, ideal-statevector quantum kernels, controlled binomial shot emulation, offline policy evaluation on frozen outputs and a local executable contract.

Version 1.3.7 adds only the preregistered corrective closure: overflow-inclusive shared histogram edges repair the two JSD drift coordinates, and the existing label interventions are scored under the same clean-resample geometry as the calibration draws. The datasets, environments, splits, models, seeds, attacks, strengths, batch sizes, policies and structural statements are unchanged; all earlier evidence directories remain immutable and the new evidence is manifested separately.

2 FORMAL CORE WITH PROOFS

2.1 Objects

A workflow state consists of the primitive (stored) artifacts $s = (X, P, C, E, \xi, g, f, y) \in \mathcal{S}$ and of the derived artifacts the workflow recomputes from them: $\tilde{X} = P(X)$, the semantic kernel $K_{\text{sem}} = \Phi(C)$ on a fixed probe set, the finite-shot estimate $\hat{K} = \text{est}(K_{\text{sem}}, E, \xi)$ with estimation randomness ξ , the observed kernel $K_{\text{obs}} = g(\hat{K})$ delivered downstream after post-processing g , the predictions $\hat{y} = f(\tilde{X}, K_{\text{obs}})$

indexed by item identity $i = 1, \dots, n$, and $R = r(\hat{y}, y)$. An intervention a overwrites a declared set of nodes (primitive or derived), keeps every node that is not a descendant of an overwritten node fixed, and recomputes the descendants of the overwritten nodes through the workflow with ξ retained; the baseline is s_0 and $s_a = a(s_0)$. Classes: T_y (label-only: overwrites y ; only R is recomputed), $T_y^\pi \subset T_y$ (prior-preserving), the feature-side mechanisms (overwrite $\tilde{X}$ or X ; $\hat{y}$ and R recomputed), including the adaptive cluster-preserving variants of Gate A, and, on the quantum branch, circuit-side (overwrite C), estimation-variation (overwrite ξ or E) and post-processing (overwrite g or K_{obs}) interventions. The executable counterpart of these semantics is `src/integrity/workflow_state.py`. The conclusion is $R(s)$ (balanced accuracy) and $\Delta_R(a) = R(s_0) - R(s_a)$; a is material at level τ if $|\Delta_R(a)| \geq \tau$. The limit $\tau \rightarrow 0^+$ is a structural-sensitivity endpoint (any conclusion change), not an operational risk threshold; $\tau = 0.02$ and 0.05 are the fixed larger-effect sensitivity analyses. An information set $\mathcal{I}$ is a view $V_{\mathcal{I}} : \mathcal{S} \rightarrow \mathcal{O}_{\mathcal{I}}$; refinement $\mathcal{I} \sqsubseteq \mathcal{I}'$ means $V_{\mathcal{I}} = \pi \circ V_{\mathcal{I}'}$; the join $\mathcal{I} \vee \mathcal{W}$ is the pair of views. A reference is stored information about the baseline that the auditor compares the audited batch with; it is trusted when no intervention of the class can alter it. We describe references on three separate axes: provenance (historical, benchmark-protected, or deployed authenticated), granularity (aggregate or item-aligned), and decision (statistical or exact). The compact classes are not a one-dimensional trust scale. Class/Level A denotes statistically thresholded aggregate comparison and may use a historical population or a clean version of the same item set. The executed A sensors use the latter as a benchmark-protected oracle without item-wise pairing; no deployed authentication mechanism is demonstrated. Level B is a deployed-trusted aggregate same-batch reference $\rho = V_{\mathcal{J}}(s_0)$ with $V_{\mathcal{J}}$ permutation-invariant; Level C is a deployed-trusted item-aligned same-batch reference with $V_{\mathcal{J}}$ item-indexed. Thus same items in a benchmark computation does not imply a trusted deployed anchor. Every selected batch sensor in the executed policy belongs to Class A; the sensor-level audit below records the exact current and reference values.

2.2 Statements and Proofs

Lemma 1 (structural blindness). If $V_{\mathcal{I}}(s_a) = V_{\mathcal{I}}(s_0)$ then $S(s_a) = S(s_0)$ for every sensor S admissible in $\mathcal{I}$.

Proof: $S(s_a) = S(V_{\mathcal{I}}(s_a)) = S(V_{\mathcal{I}}(s_0)) = S(s_0)$. For a randomized sensor whose seed is part of the view the identity holds path-wise; with a fresh seed the two outputs are equal in distribution. $\square$

Definition (blind regions and gaps). $B_{\mathcal{I}}(\mathcal{A}) = \{a \in \mathcal{A} : V_{\mathcal{I}}(a(s_0)) = V_{\mathcal{I}}(s_0)\}$; $B_{\mathcal{I},S}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0) \forall S \in \mathcal{S}\}$; $G_{\mathcal{I}}(\tau) = \{a : |\Delta_R(a)| \geq \tau, a \in B_{\mathcal{I}}\}$ and $G_{\mathcal{I},S}(\tau)$ analogously. By Lemma 1, $B_{\mathcal{I}} \subseteq B_{\mathcal{I},S}$ and $G_{\mathcal{I}} \subseteq G_{\mathcal{I},S}$, with equality iff $\mathcal{S}$ is baseline-separating on the orbit of $\mathcal{A}$: it takes a different value at $a(s_0)$ than at s_0 whenever the views differ. Pairwise separation of intervened states (injectivity on the orbit), which identification of the intervention would need, is never assumed.

Proposition 1 (monotonicity). If $\mathcal{I} \subseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_{\mathcal{I}}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_{\mathcal{I}}(\tau)$.

Proof: If $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ then $V_{\mathcal{I}}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_{\mathcal{I}}(s_0)$. Membership in the gap adds only the materiality condition, which does not depend on the view. $\square$

Corollary 1. With f fixed and deterministic: (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$ and every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change; (b) $T_y^\pi \subseteq B_{\mathcal{I}_{Ym}}$ and every sensor admissible in $\mathcal{I}_{Ym}$ is invariant under a prior-preserving change.

Proof: (a) T_y fixes the primitives X, P, C, E, ξ, g and f . Therefore $\tilde{X}, K_{\text{sem}}, \hat{K}$ and K_{obs} remain fixed by their declared dependencies, and so does $\hat{y} = f(\tilde{X}, K_{\text{obs}})$; the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$ is unchanged. Apply Lemma 1 and Proposition 1 with $\mathcal{I}_X \subseteq \mathcal{I}_{XF}$. (b) T_y^π preserves the class histogram by definition. $\square$

Proposition 2 (closure). $B_{\mathcal{I} \vee \mathcal{W}}(\mathcal{A}) = B_{\mathcal{I}}(\mathcal{A}) \cap B_{\mathcal{W}}(\mathcal{A})$; a class $\mathcal{A} \subseteq B_{\mathcal{I}}$ is completely separable in $\mathcal{I} \vee \mathcal{W}$ iff $V_{\mathcal{W}}(a(s_0)) \neq V_{\mathcal{W}}(s_0)$ for every $a \in \mathcal{A}$.

Proof: The joint view equals the baseline joint view iff both components do. Necessity: an a that leaves $V_{\mathcal{W}}$ unchanged stays in the joint blind region. Sufficiency: a changed component changes the pair. $\square$

Corollary 2. With f fixed and deterministic, every view factoring through $(\tilde{X}, \hat{y}, \text{hist}(y))$ leaves every $a \in T_y^\pi$ in its blind region. The multiset $\mathcal{I}_{XFY}$ separates $a \in T_y^\pi$ unless a permutes labels only among items with identical $(\tilde{x}, \hat{y})$; an item-aligned reference $\mathcal{I}_{XFY}^*$ separates every non-identity relabeling.

Proof: Apply Proposition 2 with $\mathcal{W}$ the view in question: by Corollary 1 each of the three components is invariant, so their join is. For the multiset of triples, two states have the same multiset iff a permutation of items maps one to the other; since $(\tilde{x}_i, \hat{y}_i)$ is fixed, such a permutation can only move labels among items with identical $(\tilde{x}, \hat{y})$. An item-indexed tuple has no such freedom. $\square$

Proposition 3 (materiality forces aggregate separability). If $R = g(M)$ with M the confusion matrix, $a \in T_y$ and $\Delta_R(a) \neq 0$, then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for $\tau > 0$.

Proof: $M(s_a) = M(s_0)$ implies $R(s_a) = g(M(s_a)) = g(M(s_0)) = R(s_0)$. The confusion matrix is a function of

the multiset of triples, so a changed M implies a changed multiset. $\square$

Remark (other metrics). The proof uses only $R = g(M)$, so the statement holds for accuracy, precision, recall, F_1 and every function of the binarised confusion matrix, with M as the sufficient aggregate. ROC-AUC is a function of the multiset of (score, label) pairs and not of M : two label vectors with the same confusion matrix can have different AUC. The same argument then holds with that multiset as the aggregate: a label-only change that changes the AUC changes the score-label multiset, a trusted stored copy of it detects every material change exactly, and item identity again needs an item-aligned reference. Both facts are checked by enumeration in the tests of the artifact; no AUC experiment is run.

Proposition 4 (separability and anchoring). (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$ (including a seed carried in the view), $a \in B_{\mathcal{I}}$ implies $\text{fire}(s_a) = \text{fire}(s_0)$ pathwise under the same state and reference construction. This pathwise equality does not in general imply equality with a false-action rate estimated under a different clean-resampling or reference construction. (ii) A reference-anchored auditor with trusted $\rho = V_{\mathcal{J}}(s_0)$ (level B or C) and rule “fire iff $d(V_{\mathcal{J}}(s), \rho) > 0$ ” has zero false-alarm probability and detects every $a \notin B_{\mathcal{J}}$ with probability one.

Proof: (i) is Lemma 1 applied to the decision function. (ii) $d(\rho, \rho) = 0$ and $d(V_{\mathcal{J}}(s_a), \rho) > 0$ iff $V_{\mathcal{J}}(s_a) \neq \rho$, which is $a \notin B_{\mathcal{J}}$ because ρ is unaltered. $\square$

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed and deterministic. (a) A trusted exact same-batch claim reference $R_0 = R(s_0)$ with “fire iff $R(s_a) \neq R_0$ ” detects exactly every conclusion change and suffices for a conclusion-only claim. A trusted aggregate reference $M_0 = M(s_0)$ detects every violation of aggregate integrity and, by Proposition 3, every material conclusion change. (b) Any reference that factors through $M, \text{hist}(y)$ or R is invariant under relabelings that permute labels among items with equal predictions (counterexample C1), which violate item-identity integrity; an item-aligned reference y_0 detects every non-identity relabeling.

Proof: (a) Proposition 4(ii) with $\mathcal{J}$ the conclusion view gives the exact R_0 comparison; applying it with the confusion view gives the exact M_0 comparison, and Proposition 3 supplies the material-conclusion consequence. (b) Swapping y_i, y_j with $\hat{y}_i = \hat{y}_j$ moves one item from cell $(\hat{y}_i, y_i)$ to $(\hat{y}_i, y_j)$ and another from $(\hat{y}_j, y_j)$ to $(\hat{y}_j, y_i)$; the moves cancel in M , hence in every function of M , and the histogram is unchanged, whereas the item-indexed tuple differs in positions i and j . $\square$

Proposition 5 (union of calibrated sensors versus conformal family calibration). (a) Let $E_1, \dots, E_m$ be the null firing events of m sensors with $p_j = \Pr_0[E_j] \leq \alpha$. Then

$$\max_j p_j \leq \Pr_0 \left[\bigcup_j E_j \right] \leq \min \left(1, \sum_j p_j \right) \leq \min(1, m\alpha). \quad (1)$$

Without assuming that some $p_j = \alpha$ the universal lower bound is $\max_j p_j$, which can be 0. If every $p_j = \alpha$, then $\alpha \leq \Pr_0[\bigcup_j E_j] \leq \min(1, m\alpha)$: the lower bound is attained by nested (coincident) events, the upper bound $m\alpha$ (when $m\alpha \leq 1$) by mutually disjoint events, and under indepen-

dence $\Pr_0[\bigcup_j E_j] = 1 - \prod_j (1 - p_j) = 1 - (1 - \alpha)^m$, strictly between the two bounds for $m \geq 2$ and $0 < \alpha < 1$.

(b) Let $v_1, \dots, v_n \in \mathbb{R}^m$ be the sensor vectors of the calibration draws and v_{n+1} that of the audited batch. Give every member j of the augmented set the leave-one-out family score

$$\tilde{U}_j = \max_{s \leq m} \frac{\#\{i \neq j : v_{s,i} < v_{s,j}\}}{n}, \quad (2)$$

computed against the other n members, and define $p = (1 + \#\{i \leq n : \tilde{U}_i \geq \tilde{U}_{n+1}\})/(n+1)$; the family fires iff $p \leq \alpha$, ties counting against firing. Let $k = \lfloor \alpha(n+1) \rfloor$ and $N_j = \#\{i \leq n+1 : \tilde{U}_i \geq \tilde{U}_j\}$. Then (i) for every augmented set at most k members satisfy $N_j \leq k$; (ii) if the $n+1$ vectors are exchangeable,

$$\Pr[p \leq \alpha] \leq \frac{\lfloor \alpha(n+1) \rfloor}{n+1} \leq \alpha, \quad (3)$$

with equality in the first inequality when the scores are almost surely distinct, and with no assumption on ties or continuity; (iii) for a single sensor without ties the rule coincides with “fire iff v_{n+1} exceeds the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest calibration value”; (iv) the level is marginal over the joint draw of calibration set and audited batch, audited batches sharing one calibration set are dependent, and nothing is claimed when exchangeability fails. For $n = 200$, $\alpha = 0.05$: $k = 10$ and the level is $10/201 = 0.0498$.

(c) An asymmetric construction scores calibration draws against the other $n-1$ calibration draws only, the audited batch against all n , and fired when the audited score exceeded the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest calibration score. It lacks the claimed level $\alpha + 1/(n+1)$. For $n = 4$, $\alpha = 0.2$ and three sensors with cyclic orderings $s_1 : j_1 > j_2 > j_3 > x > y$, $s_2 : j_2 > j_3 > j_1 > x > y$, $s_3 : j_3 > j_1 > j_2 > x > y$, the members j_1, j_2, j_3 each fire when audited (calibration scores 0.75, 0.75, 0.25, 0, threshold 0.75, audited score 1), so the firing probability under exchangeability is 0.60 against the claimed 0.40 and against α . On the frozen null draws the rule exceeds α under exchangeable re-splits (Table 16).

Proof: (a) The union contains each E_j (lower bound); Boole’s inequality gives the upper bound; nested, disjoint and independent events attain or realise the stated values, the last by inclusion–exclusion. (b)(i) Suppose $k+1$ members satisfy $N_j \leq k$ and let j^* be one with the smallest score among them; every one of the $k+1$ has $\tilde{U} \geq \tilde{U}_{j^*}$, so $N_{j^*} \geq k+1$, a contradiction. (ii) The score of member j depends on v_j and on the multiset of the others, so the score map is permutation-equivariant and exchangeability of the v ’s gives exchangeability of the $\tilde{U}$ ’s and of the N_j ’s; $p \leq \alpha$ iff $N_{n+1} \leq k$, and by exchangeability $\Pr[N_{n+1} \leq k] = \mathbb{E}[\#\{j : N_j \leq k\}]/(n+1) \leq k/(n+1)$ by (i). With distinct scores exactly k members satisfy $N_j \leq k$; ties merge ranks and shrink that set. (iii) For $m = 1$ and distinct values, $\tilde{U}_i \geq \tilde{U}_{n+1}$ iff $v_i > v_{n+1}$, so $p = (1 + \#\{i : v_i > v_{n+1}\})/(n+1) \leq \alpha$ iff v_{n+1} exceeds at least $n - k + 1 = \lceil (n+1)(1-\alpha) \rceil$ calibration values. (iv) restates what (ii) proves. (c) is checked by enumeration in the artifact tests, which also verify (i) over every weak ordering (all tie patterns) of up to seven elements with one sensor, five with two sensors and four with three sensors, and over random tied configurations with up to ten sensors. $\square$

Proposition 6 (no local authentication without an uncontrolled root). Let a *local verifier* be any map $\text{acc} : \mathcal{O}_{\mathcal{I}} \times \text{Ref} \rightarrow \{\text{accept}, \text{reject}\}$ of the observed view and a reference bundle, let $\mathcal{S}_H$ be the honest family of states and $\rho_H(s) = V_{\mathcal{I}}(s)$ the reference the honest process attaches to s . Assume (honest completeness) $\text{acc}(V_{\mathcal{I}}(s), \rho_H(s)) = \text{accept}$ for every $s \in \mathcal{S}_H$, and (joint control) the adversarial class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some $s' \in \mathcal{S}_H$ with $s' \not\sim_{\mathcal{I}} s_0$. Then the verifier accepts the substituted pair, which is observationally an honest run of s' , so no function of that input establishes authenticity of the served state relative to s_0 ; if $R(s') \neq R(s_0)$ a material change is served. Conversely, if a component $\rho^* = V_{\mathcal{K}}(s_0)$ of the bundle cannot be written by the class, “reject iff $V_{\mathcal{K}}(s) \neq \rho^*$ ” rejects every substitution with $V_{\mathcal{K}}(s') \neq V_{\mathcal{K}}(s_0)$.

Proof: The substituted input equals the honest input of s' ; by honest completeness it is accepted, and acceptance is the same event as for an honest run of s' , so no function of the input distinguishes the two. The converse is Proposition 4(ii) applied to $\mathcal{K}$. The statement concerns authenticity, not consistency: by Proposition 3 a material substitution changes the joint view, but the change is invisible to a verifier all of whose evidence the class can rewrite. $\square$

Proposition 7 (the quantum branch as an instance of the lattice). Let $\mathcal{C}$ be the circuit representations, $\Phi : \mathcal{C} \rightarrow \mathcal{K}$ the semantic kernel map on a fixed probe set, $K_{\text{sem}} = \Phi(\mathcal{C})$, $\hat{K}$ the finite-shot estimate, K_{obs} the observed kernel, h the provenance hash, A the algebraic invariants and f_K the decision from a kernel. Three intervention classes are distinguished: (a) circuit-side (overwrites $\mathcal{C}$; may change K_{sem}); (b) estimation variation ($\mathcal{C}$ and K_{sem} fixed; $\hat{K}$ changes); (c) post-processing or kernel substitution ($\mathcal{C}$, K_{sem} and $\hat{K}$ fixed; K_{obs} changes). Assume h collision-free on the circuits under study. (i) On class (a), $[\mathcal{C}] = \Phi^{-1}(\Phi(\mathcal{C}))$ is the semantic class of $\mathcal{C}$; approved rewrites map $\mathcal{C}$ into $[\mathcal{C}]$ without fixing $h(\mathcal{C})$, so V_h refines $V_{K_{\text{sem}}}$ and $B_h(\mathcal{A}) \subseteq B_{K_{\text{sem}}}(\mathcal{A})$, strictly whenever $\mathcal{A}$ contains an approved rewrite. (ii) On class (c), $V_A = A \circ V_{K_{\text{obs}}}$ and $V_f = f(\tilde{X}) \circ V_{K_{\text{obs}}}$, so $B_{K_{\text{obs}}} \subseteq B_A$ and $B_{K_{\text{obs}}} \subseteq B_f$; a PSD-preserving substitution $K' \neq K_{\text{obs},0}$ with $A(K') = A(K_{\text{obs},0})$ lies in $B_A \setminus B_{K_{\text{obs}}}$ and, because $\mathcal{C}$ and K_{sem} are fixed on this class, also in $B_h \cap B_{K_{\text{sem}}}$: it is closed only by a trusted reference on K_{obs} itself, so the trusted semantic probe of $\mathcal{C}$ and the trusted observed-kernel reference are different anchors. A kernel change that crosses no decision boundary lies in $B_f \setminus B_{K_{\text{obs}}}$. (iii) On class (b), for an honest finite-shot estimate the acceptance rule “accept iff $\hat{K} = K_{\text{sem},0}$ ” has false-alarm probability $1 - \Pr[\hat{K} = K_{\text{sem},0} \mid \text{honest}]$, which depends on the discrete support of the estimator and may be large or equal to one but is not universally one; exact equality is therefore not an appropriate acceptance criterion under finite-shot estimation, $d(\hat{K}, K_{\text{sem},0})$ must be treated as a level-A statistic with a null from repeated honest estimation, to which Proposition 5(b) applies with $m = 1$ when the repeated estimates and the audited one are exchangeable, and where no such null is calibrated no statistical integrity claim is made. No inclusion is claimed across classes.

Proof: (i) If $h(C_1) = h(C_2)$ then $C_1 = C_2$ by collision freedom and $\Phi(C_1) = \Phi(C_2)$; hence V_h refines $V_{K_{\text{sem}}}$ and Proposition 1 gives the inclusion; an approved rewrite

TABLE 1

Counterexample witnesses realised in the frozen expansion (count / denominator). W1–W4 are evaluation-label rows, W5–W6 feature-side rows, W7 all intervened rows, W8 material label rows; W9 and W10 count the label rows that a trusted aggregate reference, respectively only an item-aligned reference, detects (Corollary 3).

ID	Property	Count
W1	identity change without joint-outcome or conclusion change	808 / 3,600
W2	conclusion change without model-output change	2,617 / 3,600
W3	marginal invariance with conclusion change	1,059 / 1,800
W4	confusion-profile change without conclusion change	175 / 3,600
W5	model-output change without conclusion change	296 / 7,200
W6	representation change without output change	2,513 / 7,200
W7	apparent improvement under intervention	1,534 / 10,800
W8	material conclusion change always changes the confusion profile (label path)	2,617 / 2,617
W9	label rows detectable by a trusted aggregate (confusion-matrix) reference of the same batch	2,792 / 3,600
W10	label rows detectable only by an item-aligned reference	808 / 3,600

$C' \in [C_0]$ with a different canonical text has $h(C') \neq h(C_0)$ and $\Phi(C') = \Phi(C_0)$. (ii) is Proposition 1 applied to the two coarsenings of $V_{K_{\text{obs}}}$; on class (c) the primitive C is untouched, so $h(C)$ and $\Phi(C)$ are unchanged by the definition of an intervention (Section 2.1), which gives the membership in $B_h \cap B_{K_{\text{sem}}}$. The set differences are witnessed in Table 7 (15/15 PSD-preserving substitutions with silent algebraic sensors and unchanged circuit hash, exposed by the anchored comparison of the delivered kernel; RZ mutations at 0.02 with zero output change and firing semantic sensor). (iii) For an entry with fidelity p estimated from N shots, $N\hat{p} \sim \text{Binomial}(N, p)$ and $\Pr[\hat{p} = p] = \binom{N}{Np} p^{Np} (1-p)^{N-Np}$ when Np is an integer and 0 otherwise; for $p = 1/2$ and $N = 2$ this is $1/2$, so the false-alarm probability of exact equality is $1/2$, not one, and for $p = 1/3$, $N = 2$ it is one. For a kernel of independently estimated entries the probabilities multiply. The universal statement “ $\Pr[\hat{K} = K_{\text{sem},0}] = 0$ under non-degenerate shot noise” is false on the support of the estimator; the conclusion that exact equality is not an acceptance criterion and that $d(\hat{K}, K_{\text{sem},0})$ needs a calibrated null stands, with the rest being the definition of the batch-level auditor and Proposition 5(b). The arithmetic is checked in `tests/test_workflow_state.py`. $\square$

2.3 Counterexamples

Table 1 restates the witnesses of the main text. C1 is the swap of two labels between items with equal predictions; C3 between items with different predictions; C4 uses two compensating flips that move the same number of items across the diagonal of M in each class; C5 and C6 are feature-side; C7 counts every intervened row whose balanced accuracy increased; C8 is Proposition 3; W9 and W10 count the label rows that a trusted aggregate reference of the same batch detects (every material one) and those that only an item-aligned reference exposes (Corollary 3). The cluster-preserving drift of Gate A (C9 in the artifact’s formal core) is not a structural blind region but a concrete stress test of sensor insufficiency against an adaptive attacker. It is measured in Section 12, not proved or presented as priority for generic monitor-aware drift evasion.

3 ADVERSARY MODEL: READING RULE

Every executed intervention is assigned to one of four classes: *fault robustness* (benign or accidental change), *integrity corruption* (non-adaptive change by a party that does not model the auditor), *adversarial attack* (targeted change to alter the reported conclusion) and *adaptive attacker* (targeted change designed with knowledge of the monitored statistics). The complete per-class record (actor, capability, access point, knowledge, objective, budget, alterable assets, roots the class cannot write, observable information set and supported claim) is `manuscript/ADVERSARY_MODEL.md` in the artifact; the main article prints the summary table. The trusted computing base is: the local runtime and host; the training data, training routine and fitted model; the reference input array and preprocessing declaration; the reference circuit and probe kernels; SHA-256 collision resistance; the item-level label reference only where $\mathcal{I}_{XY}^*$ is declared; and the code that builds and verifies the envelope.

4 FROZEN EXPERIMENTAL DESIGN

4.1 Expansion Environments

The expansion comprises eight fixed environments (Table 2): five from CICIDS2017, two from UNSW-NB15 and one from ToN-IoT; with Gate 1, six of the nine environments of the study are CICIDS2017. “OOD” means a prespecified source-to-target construction, not a sampled deployment population. Every environment uses five split seeds and two nested model seeds and evaluates projected dimensions 8, 10 and 12. The CICIDS scale gate uses 256/256 train/evaluation samples; all others use 128/128.

Each staged table contains 3,000 balanced binary rows before experimental subsampling. Numeric non-finite values are replaced during staging; categorical fields are excluded. Several temporal targets yield clean performance near chance (Table 3), limiting the magnitude that later interventions can express; the exact information-set invariances do not depend on clean headroom. The table is a presentation-only aggregation of frozen observations, not a new inferential analysis.

4.2 Perturbation Suite

The paper-core suite crosses six mechanisms with nominal strengths 0.02, 0.05 and 0.10 (Table 4). Artifact identifiers retain the historical family string `target_shift`; the scientific text uses *evaluation-label intervention* because these rows do not instantiate the statistical label-shift problem. Gate A adds cluster-preserving variants of mechanisms 4–9 at strengths 0.02, 0.05, 0.10, 0.25 and 0.50 as a concrete monitor-aware stress test of the declared fingerprint (Section 12).

5 INFORMATION-SET COVERAGE CONTRACT

Table 6 records what each evidence regime can establish in the implemented study. Structural blindness means that the view is unchanged by construction; it is stronger and narrower than observing zero empirical response in a finite sample, and both are distinct from calibrated detectability and from response reduction against a declared monitored fingerprint.

TABLE 2
Prespecified expansion environments.

ID	Environment	Training source	Evaluation source	Samples	Profiles
E1	CICIDS scale/ID	Frozen CICIDS subset	Same staged source	256/256	SVC, ZZ
E2	UNSW-NB15 ID	Balanced staged UNSW	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ- equivalent
E3	ToN-IoT ID	Balanced staged ToN-IoT	Same staged source	128/128	SVC, ZZ, PauliXYZ, Z/PauliXZ- equivalent
E4	CICIDS Tue→Wed	Tuesday traffic	Wednesday traffic	128/128	SVC, ZZ
E5	CICIDS Tue→Fri-PortScan	Tuesday traffic	Friday PortScan traffic	128/128	SVC, ZZ
E6	CICIDS Wed→Thu-Web	Wednesday traffic	Thursday Web traffic	128/128	SVC, ZZ
E7	CICIDS Wed→Fri-AM	Wednesday traffic	Friday morning traffic	128/128	SVC, ZZ
E8	UNSW temporal OOD	Prespecified UNSW source period	Prespecified UNSW target period	128/128	SVC, ZZ

TABLE 3

Clean balanced accuracy by expansion environment. Values are mean [range] over five split clusters after averaging the prespecified dimension/model/model-seed cells within each cluster.

Environment	Clean BA mean [range]
E1 CICIDS scale/ID	0.762 [0.756, 0.766]
E2 UNSW-NB15 ID	0.766 [0.738, 0.800]
E3 ToN-IoT ID	0.752 [0.716, 0.807]
E4 CICIDS Tue→Wed	0.492 [0.480, 0.520]
E5 CICIDS Tue→Fri-PortScan	0.590 [0.565, 0.615]
E6 CICIDS Wed→Thu-Web	0.483 [0.454, 0.518]
E7 CICIDS Wed→Fri-AM	0.459 [0.431, 0.493]
E8 UNSW temporal OOD	0.759 [0.743, 0.775]

TABLE 4

Frozen 18-condition intervention suite.

IDs	Mechanism	Strengths
1–3	Feature sign flip	0.02, 0.05, 0.10
4–6	Feature-wise mean shift	0.02, 0.05, 0.10
7–9	Scaling drift	0.02, 0.05, 0.10
10–12	Feature dropout	0.02, 0.05, 0.10
13–15	Prior-preserving evaluation-label flip	0.02, 0.05, 0.10
16–18	Random evaluation-label flip	0.02, 0.05, 0.10

The implementation chain is explicit. The runner constructs feature, prediction, label-marginal and confusion-profile scores from the current batch and the clean benchmark-held arrays; the statistics consume aggregate distributions and not item correspondence. Calibration converts those scores to family flags. Gate D and `src/hsaas/policy.py` consume only the union flag, family flags and exact-reference flag—never raw `X_te_f`, `y_te_f`, or a clean batch. Producing the selected Class-A scores upstream nevertheless requires the benchmark oracle shown in Table 5. The benchmark protects that oracle from its executed attack API, but the artifact neither authenticates nor delivers it to a deployed auditor. By contrast, Class B/C exact flags presuppose a deployed-authenticated reference at the declared granularity.

Every machine-readable coverage record contains eight fields: workflow boundary; protected asset; adversary or failure class including fixed elements; available information

regime and trusted references; sensor, calibration rule, its level and premise; covered perturbations and structural, calibrated and adaptive residual blind regions; interruption on the prespecified near-null stress controls; response or countermeasure.

6 DEDUPLICATION AND STATISTICAL UNITS

Gate 1 emits 7,980 raw model-intervention rows; different quantum-map jobs repeat the same classical SVC pipeline, producing 3,420 repeated rows that are accepted as duplicates only after numerical identity checks, leaving 4,560 unique observations. The expansion emits 13,680 raw rows from 360 completed CSV/JSON job pairs with 2,280 verified repeats, leaving 11,400 unique observations. The strict builder fails if a job is missing, an environment contains an unexpected design, or a duplicated row disagrees.

Coverage counts use prespecified cells as denominators. For the secondary suite-average model profile, nested model-seed values are averaged inside each of five split seeds; the split seed is the inferential unit and the interval is $\bar{d} \pm t_{0.975,4} s_d / \sqrt{5}$. For the calibration gates the dependence structure is explicit: each of the 240 runs (eight environments $\times$ five split seeds $\times$ two model seeds $\times$ three dimensions) draws 20 calibration and 20 evaluation batches as overlapping simple random subsets of one pool half, the 200 draws of a cell come from ten pool halves, and the 200 evaluation draws of a cell share one calibration set; the inferential unit is the (environment, split-seed) cluster (40 clusters, five per environment). Pooled rates over draws and the Clopper–Pearson intervals of artifact 1.1.0 are descriptive. The conformal level of Proposition 5(b) is marginal, not conditional on the calibration set. No multiplicity adjustment or population transport is claimed.

7 EXACT-STATEVECTOR ENGINE VALIDATION

The expansion evaluator replaces pairwise compute-uncompute execution with a mathematically equivalent ideal-statevector calculation: it prepares each unique state $|\phi(x)\rangle$ once and forms $K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$ by matrix multiplication, with the same eigenvalue-clipping PSD repair as the frozen Qiskit reference. Validation has four layers:

TABLE 5

Audited reference semantics of every selected sensor in v1.3.2. Pairing means that item correspondence enters the statistic; same set means the runner uses the clean version of the audited item set. Harness protected means that the benchmark keeps the reference outside the executed attack API; trusted means that the threat model grants deployed authentication. Hist./stat. reports historical and statistical reference semantics in that order. Class A is statistical comparison; B and C are trusted aggregate and trusted item-aligned exact invariants.

Sensor	Current value	Reference value	Item pair	Same set	Harness protected	Threat-model trusted	Hist. / stat.	Rule / class
integrity_jsd_vs_clean_eval	empirical feature distributions of X_{eval}	distribution of X_{te_f} clean feature distributions	no	yes	yes	no	no / yes	statistical / A
integrity_mmd_vs_clean_eval	feature multiset X_{eval}	feature multiset X_{te_f}	no	yes	yes	no	no / yes	statistical / A
integrity_ks_reject05_vs_clean_eval	per-feature distributions of X_{eval}	per-feature distributions of X_{te_f}	no	yes	yes	no	no / yes	statistical / A
integrity_score_jsd_vs_clean_eval	score distribution $f(X_{eval})$	score distribution $f(X_{te_f})$	no	yes	yes	no	no / yes	statistical / A
integrity_pred_pos_rate_shift	mean predicted-positive rate on X_{eval}	mean predicted-positive rate on X_{te_f}	no	yes	yes	no	no / yes	statistical / A
integrity_pred_jsd	predicted-class distribution on X_{eval}	predicted-class distribution on X_{te_f}	no	yes	yes	no	no / yes	statistical / A
integrity_label_prior_shift	mean positive-label rate of y_{eval}	mean positive-label rate of y_{te_f}	no	yes	yes	no	no / yes	statistical / A
integrity_label_jsd	binary label distribution of y_{eval}	binary label distribution of y_{te_f}	no	yes	yes	no	no / yes	statistical / A
integrity_confusion_profile_ll	aggregate confusion PMF of $(y_{eval}, f(X_{eval}))$	aggregate confusion PMF of $(y_{te_f}, f(X_{te_f}))$	no	yes	yes	no	no / yes	statistical / A
integrity_confusion_profile_jsd	aggregate confusion PMF of $(y_{eval}, f(X_{eval}))$	aggregate confusion PMF of $(y_{te_f}, f(X_{te_f}))$	no	yes	yes	no	no / yes	statistical / A
integrity_confusion_profile_ll_delta	aggregate confusion-profile L1 delta	trusted aggregate confusion profile of the same batch	no	yes	yes	yes	no / no	exact / B
integrity_confusion_profile_jsd_delta	aggregate confusion-profile JSD delta	trusted aggregate confusion profile of the same batch	no	yes	yes	yes	no / no	exact / B
integrity_pred_disagreement	item-indexed predictions $f(X_{eval})$	trusted item-indexed predictions $f(X_{te_f})$	yes	yes	yes	yes	no / no	exact / C
label_flip_rate	item-indexed y_{eval}	trusted item-indexed labels y_{te_f}	yes	yes	yes	yes	no / no	exact / C

TABLE 6

Evidence regimes, implemented sensors, demonstrated blind regions and calibrated status (conformal family rule).

Regime	Evidence	Implemented sensor family	Structural blind region	Calibrated / exact status
$\mathcal{I}_X$	Features	Feature JSD, MMD, KS rejection	Every label-only change	Conformal decision FPR 0.056; label path 0/3,600; evaded by cluster-preserving drift at matched strengths
$\mathcal{I}_{XF}$	Features, scores, predictions	$\mathcal{I}_X$ plus score JSD, prediction-rate shift, prediction JSD	Every label-only change for a fixed predictor	Conformal FPR 0.059; label path 0/3,600; partially evaded by cluster-preserving drift
$\mathcal{I}_{Y_m}$	Label class counts	Prior delta and label JSD	Prior-preserving relabeling; every feature-side change	FPR 0.048; zero containment; 5,450 residual-blind material cases at $\tau \rightarrow 0^+$
$\mathcal{I}_{XFY}$ (batch)	Item triples; Class-A statistical comparison	Confusion-profile statistics added	None for material label changes (Proposition 3), but finite-batch misses under the frozen rules: 11–43 of 2,617	Conformal FPR 0.055
$\mathcal{I}_{XFY}^*$	Deployed-trusted same-batch references (B aggregate; C item-aligned)	Aggregate confusion-profile deltas; item-aligned prediction disagreement and label mismatch (exact-zero null)	None	Exact: 2,617/2,617 material label rows and every material adaptive row; 0 material corruptions served; 0 invariant violations on 1,200 exact-zero rows; 85 gross exact blocks and 41 net additional near-null interruptions versus batch $\mathcal{I}_{XFY}/P2$
$\mathcal{I}_Q$	Circuit, parameters, kernel, estimation evidence	Hashes, semantic probes, algebra, repeated estimation, output changes	Composition-dependent: hashes overreact to equivalence; algebra misses PSD-preserving substitution; outputs miss sub-decision changes (Proposition 7)	165/165 cells; nine acceptance checks

TABLE 7
Quantum-gate sensor response fractions.

Condition	Circuit prov.	Kernel prov.	Anchored kernel comparison	Algebraic kernel	Repeated estimate	Output change
Clean	0.00	0.00	0.00	0.00	0.00	0.00
Benign transpilation	1.00	1.00	0.00	0.00	0.00	0.00
Common-unitary rewrite	1.00	1.00	0.00	0.00	0.00	0.00
Parameterized RZ 0.02	1.00	1.00	1.00	0.00	0.00	0.00
Parameterized RZ 0.10	1.00	1.00	1.00	0.00	0.00	0.13
Feature-map reps 1→2	1.00	1.00	1.00	0.00	0.00	0.73
Asymmetric kernel edit	0.00	1.00	1.00	1.00	0.00	0.00
Kernel diagonal erosion	0.00	1.00	1.00	1.00	0.00	0.00
PSD-preserving kernel mix	0.00	1.00	1.00	0.00	0.00	0.07
Shot estimator 256	0.00	1.00	1.00	1.00	1.00	0.47
Shot estimator 1,024	0.00	1.00	1.00	1.00	1.00	0.20

symmetric and cross-kernel unit matrices agree with the reference at relative and absolute tolerance 10^{-10} ; symmetry, unit diagonal, numerical range, cache bounds and cache reuse are unit-tested; a CICODES 128/128 end-to-end replay (split/model seed 42, dimension 8, ZZ, paper-core suite) preserves every categorical, discrete performance, impact and integrity-signal endpoint, with only ROC-AUC and its derived drops differing by at most 1.221896×10^{-4} (one half-tie increment for 64 positive and 64 negative examples); and the prespecified 256/256 pilot completes. The 240 jobs of Gate A add a fifth layer: their clean rows and matched controls reproduce the frozen expansion rows with a maximum absolute difference of 0.0 in thirteen columns over 4,200 rows. The engine is tagged `exact_statevector`; it supplies no finite-shot, noise-model or hardware evidence.

8 QUANTUM GATE: CONDITIONS AND COVERAGE

The quantum gate crosses five split seeds, dimensions 4/6/8 and 11 conditions, giving 165 design cells. Table 7 and Fig. 1 report response fractions over the 15 split-dimension cells per condition. A value of one means non-zero raw response in all 15 cells at the frozen tolerance; it is not calibrated power. The table is the empirical realisation of Proposition 7, read class by class: rows 2–3 (class (a)) change the hash and nothing semantic (i); rows 7–9 (class (c)) change the observed kernel with the circuit hash silent and are or are not visible to the algebraic coarsening (ii); rows 10–11 (class (b)) show a non-zero repeated-estimation discrepancy in every one of these cells, an empirical property of the gate that motivates the calibrated treatment of (iii), not a universal law. In this ideal-statevector gate the honest estimate equals the semantic kernel and g is the identity, so the single anchored comparison against $\Phi(C_0)$ acts as the semantic probe on class (a) and as the observed-kernel reference on class (c).

All nine frozen acceptance checks pass: 165 design cells are present; clean evidence is invariant; benign transpilation and the common-unitary rewrite alter provenance but preserve semantic kernels and outputs; every data-dependent circuit mutation changes the semantic kernel; symmetry and diagonal sensors catch their respective malformed edits; the PSD-preserving mixture passes algebra but is exposed by trusted provenance/semantics; and both shot estimators produce non-zero repeat discrepancy in every cell. The shot conditions sample exact fidelities: they exercise the contract under controlled stochastic estimation but are neither sampler-backend nor QPU runs. The algebraic check

Clean control	0.00	0.00	0.00	0.00	0.00	0.00
Benign transpilation	1.00	1.00	0.00	0.00	0.00	0.00
Common-unitary rewrite	1.00	1.00	0.00	0.00	0.00	0.00
Parameterized RZ (0.02)	1.00	1.00	1.00	0.00	0.00	0.00
Parameterized RZ (0.10)	1.00	1.00	1.00	0.00	0.00	0.13
Feature-map reps 1→2	1.00	1.00	1.00	0.00	0.00	0.73
Asymmetric kernel edit	0.00	1.00	1.00	1.00	0.00	0.00
Kernel diagonal erosion	0.00	1.00	1.00	1.00	0.00	0.00
PSD-preserving kernel mix	0.00	1.00	1.00	0.00	0.00	0.07
Shot estimator (256)	0.00	1.00	1.00	1.00	1.00	0.47
Shot estimator (1024)	0.00	1.00	1.00	1.00	1.00	0.20
	Circuit hash	Kernel hash	Semantic kernel	Algebra checks	Repeated estimate	Output change

Fig. 1. Response fraction of each quantum-branch sensor over the 15 cells (five splits, three dimensions) of every condition of the 165-design-cell simulator gate (0 = no cell responds, 1 = every cell); the values of Table 7. Circuit and kernel hashes are the provenance sensors; the anchored-kernel column compares against $\Phi(C_0)$, serving as a semantic probe for circuit-side rows and an observed-kernel reference for post-processing rows in this ideal gate. Coverage is conditional on the evidence exposed to each sensor; output invariance does not imply workflow integrity, and the shot rows are binomial emulation, not QPU runs.

concerns the raw estimated/observed kernel before any downstream PSD-repair procedure. A consumer that projects or repairs a kernel before use defines a different observable path, not evaluated here; PSD repair is containment rather than proof of integrity.

9 EXECUTABLE CONTRACT AND POLICY COMPOSITION

The prototype evaluates four ordered contracts: input/preprocessing (input array and preprocessing declaration hashes; mismatch blocks); circuit/kernel (circuit and kernel provenance, semantic probes, symmetry, diagonal, eigenvalue and PSD-repair evidence; an equivalent rewrite passes only if explicitly approved); execution/result (declared execution metadata, repeated-estimation discrepancy and prediction provenance; approved stochastic execution is held); evaluation/report (item-level labels bound to predictions and balanced accuracy recomputed; prior-preserving corruption blocks even when the marginal is identical).

TABLE 8

Composition of the frozen four-contract envelopes with the policy layer under $\mathcal{I}_{XY}^*$ (lattice maximum). Approved rewrites and approved stochastic estimation are not exact violations; the contract's own hold for the approved emulator is preserved.

Scenario	Contract	Exact viol.	Policy	Composed
Clean	allow	no	allow	allow
Approved transpilation	allow	no	allow	allow
Parameterized circuit mutation	block	yes	block	block
PSD-preserving kernel substitution	block	yes	block	block
Prior-preserving label corruption	block	yes	block	block
Approved 256-shot emulator	hold	no	allow	hold

Contracts form an ordered SHA-256 chain: later modification is detectable, but the chain is not signed and does not authenticate a provider. The contracts fail closed on their invariants: all passes yield `allow`; at least one review and no failure yields `hold`; any failure yields `block`. Six prespecified scenarios and all six chains pass eight strict acceptance checks in the frozen artifact. The policy layer composes with these contracts by the maximum in the lattice; the composition over the six frozen envelopes is Table 8, where an exact violation is derived from the recorded evidence with the approval flags respected. The policy layer is a pure function evaluated offline on frozen outputs; nothing in this article is a deployed runtime service.

10 PREREGISTERED REINFORCEMENT GATES (ARTIFACT 1.1.0)

Three sensitivity gates were added after the 1.0.0 evidence was frozen. Their protocol, endpoints and one implementation clarification (amendment A1) were recorded in `manuscript/paper15_v11_reinforcement_prereg.md` before execution; results are reported as executed. The gates reuse the exact-statevector engine, the frozen split and model seeds and the frozen 18-condition suite.

10.1 Gate N: Null Calibration with Disjoint Clean Pools

For each of the 240 expansion runs the clean pool is the set of evaluation-side rows not selected into the frozen evaluation batch ($900 - n$ in ID protocols, $3,000 - n$ in OOD protocols), transformed with the same training-fitted projection and scaler as the frozen run. A permutation seeded by the split seed divides the pool into a calibration half and an evaluation half; twenty clean batches of n rows are drawn from each half by simple random sampling without replacement; the suite also includes an identity control and two near-null controls (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$) applied in place. Ten distributional sensors are calibrated per (environment, dimension, model) cell at the 191st of 200 pooled calibration values ($\alpha = 0.05$ per sensor; fire if strictly greater); thresholds are frozen before any attacked row is examined; the empirical false-alarm rate is the firing fraction on the 200 disjoint evaluation draws. Two auditors are distinguished (amendment A1): the batch-level auditor uses only calibrated distributional sensors, and the item-aligned auditor compares the same items against trusted references with an exact-zero null. In E1 ($n = 256$) each pool

TABLE 9

Gate N: empirical false-alarm rate of the null-calibrated sensors and batch-level regimes on 12,000 disjoint clean evaluation draws (all eight environments pooled; nominal $\alpha = 0.05$ per sensor). Intervals are descriptive Clopper–Pearson 95% intervals: draws within a cell share a pool half and are not independent. Regime rows are unions without correction.

Sensor / regime	Fires	Rate	95% CI
Feature JSD	600	0.050	[0.046, 0.054]
Feature MMD	822	0.069	[0.064, 0.073]
KS rejection rate	342	0.029	[0.026, 0.032]
Score JSD	668	0.056	[0.052, 0.060]
Prediction-rate shift	520	0.043	[0.040, 0.047]
Prediction JSD	586	0.049	[0.045, 0.053]
Label prior shift	570	0.048	[0.044, 0.051]
Label JSD	570	0.048	[0.044, 0.051]
Confusion profile L_1	517	0.043	[0.040, 0.047]
Confusion profile JSD	638	0.053	[0.049, 0.057]
$\mathcal{I}_X$ (any sensor)	1500	0.125	[0.119, 0.131]
$\mathcal{I}_{XF}$ (any sensor)	2438	0.203	[0.196, 0.210]
$\mathcal{I}_{Y_m}$ (any sensor)	570	0.048	[0.044, 0.051]
$\mathcal{I}_{XFY}$ (any sensor)	3113	0.259	[0.252, 0.267]

half holds 322 rows, so the draws overlap almost entirely and E1 rates are indicative only.

Results. All 240 runs completed (14/14 acceptance checks). Table 9 reports the pooled per-sensor false-alarm rates (0.029–0.069 against the nominal 0.05) and the uncorrected regime unions; Table 10 the environment dependence; Table 11 the descriptive finite-batch detection rates under the frozen rules and the near-null control responses; Table 12 the label path (the frozen $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and prior-preserving $\mathcal{I}_{Y_m}$ fire on none of the 3,600 label observations; the batch-level $\mathcal{I}_{XFY}$ union fires on 43; the item-aligned auditor detects all 3,600); and Table 13 the cluster structure that makes the KS sensor react to in-place perturbations of order 10^{-3} standard deviations while its between-batch null stays narrow, the fingerprint that Gate A attacks. The intervals in Table 9 are descriptive.

10.2 Gates P and T: Preprocessing Ablation and Tuned Baselines

The CICIDS ID design (128/128, paper-core, ZZ and SVC, five splits, two model seeds, three dimensions) is re-executed under three scaler configurations (P0 reference: standard for SVC, min–max to 2π for QSVC; PA standard for both; PB min–max for both) and with both learners tuned on training rows only (SVC over $C \in \{0.1, 1, 10, 100\} \times \gamma \in \{\text{scale}, 0.01, 0.1, 1\}$, QSVC over $C \in \{0.1, 1, 10, 100\}$ on the precomputed training kernel, stratified five-fold cross-validation, balanced-accuracy scoring, ties toward the smallest C) in CICIDS ID and in the OOD environment with the highest mean clean SVC balanced accuracy in the frozen tables (UNSW temporal OOD, E8, 0.753). Table 14 reports the paired differences and Table 15 the clean balanced accuracies and selected constants. The direction of the secondary profile is stable (five of five positive split clusters in every configuration, dimension and environment); its magnitude is conditional: standardizing the quantum branch reduces the difference to roughly a third, matched min–max scaling leaves it unchanged, tuning widens it in CICIDS ID as ZZ gains clean headroom and leaves UNSW temporal

TABLE 10

Gate N: per-environment false-alarm rates on 1,200 evaluation draws (three dimensions, two or four models, 200 draws per cell). Sensor range is the minimum and maximum per-sensor rate in the environment; regime columns are batch-level unions. E1 draws 256 rows from 322-row pool halves and is indicative only.

Environment	n	Pool half	Sensor range	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$
E1 CICIDS ID 256	256	322	0.023–0.130	0.203	0.245	0.045	0.331
E2 UNSW-NB15 ID	128	386	0.008–0.070	0.063	0.135	0.070	0.210
E3 ToN-IoT ID	128	386	0.025–0.172	0.215	0.280	0.070	0.358
E4 Tue→Wed	128	1436	0.025–0.082	0.150	0.230	0.025	0.254
E5 Tue→Fri-PortScan	128	1436	0.018–0.051	0.066	0.160	0.025	0.198
E6 Wed→Thu-Web	128	1436	0.013–0.057	0.070	0.163	0.025	0.181
E7 Wed→Fri-AM	128	1436	0.025–0.066	0.133	0.212	0.025	0.258
E8 UNSW temporal OOD	128	1436	0.032–0.092	0.072	0.192	0.050	0.237

TABLE 11

Gate N: detection rate of each regime on the frozen `paper_core` observations (all environments, dimensions and models pooled; 600 observations per intervention row) at the null-calibrated thresholds, and firing rate on the in-place near-null controls. Batch-level regimes use calibrated distributional sensors; item-aligned regimes use exact zero thresholds on item-wise deltas.

Intervention	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	$\mathcal{I}_{XF}$ item	$\mathcal{I}_{XFY}$ item
Feature sign flip 0.02	0.118	0.192	0.000	0.200	0.793	0.793
Feature sign flip 0.05	0.532	0.640	0.000	0.643	0.867	0.867
Feature sign flip 0.10	0.757	0.877	0.000	0.878	0.907	0.907
Mean shift 0.02	1.000	1.000	0.000	1.000	0.357	0.357
Mean shift 0.05	1.000	1.000	0.000	1.000	0.500	0.500
Mean shift 0.10	1.000	1.000	0.000	1.000	0.618	0.618
Scaling drift 0.02	1.000	1.000	0.000	1.000	0.537	0.537
Scaling drift 0.05	1.000	1.000	0.000	1.000	0.608	0.608
Scaling drift 0.10	1.000	1.000	0.000	1.000	0.668	0.668
Feature dropout 0.02	0.000	0.000	0.000	0.000	0.488	0.488
Feature dropout 0.05	0.000	0.022	0.000	0.022	0.705	0.705
Feature dropout 0.10	0.000	0.062	0.000	0.062	0.763	0.763
Prior-preserving label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Prior-preserving label flip 0.05	0.000	0.000	0.000	0.002	0.000	1.000
Prior-preserving label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Random label flip 0.02	0.000	0.000	0.000	0.000	0.000	1.000
Random label flip 0.05	0.000	0.000	0.000	0.003	0.000	1.000
Random label flip 0.10	0.000	0.000	0.000	0.033	0.000	1.000
Near-null Gaussian $\sigma = 0.001$	0.713	0.713	0.000	0.713	0.042	0.042
Near-null scaling $\alpha = 0.001$	0.879	0.882	0.000	0.882	0.099	0.099

TABLE 12

Gate N: label-path interventions in the frozen expansion; number of observations on which each regime fires after calibration (batch-level) or by exact item-aligned comparison.

Subset	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{Y_m}$	$\mathcal{I}_{XFY}$	item-aligned
All label interventions	3,600	0	0	0	43	3,600
Prior-preserving	1,800	0	0	0	21	1,800
Positive conclusion impact	2,184	0	0	0	40	2,184

OOD unchanged. Exact label-path invariance holds in every configuration.

11 PREREGISTERED POLICY GATES (ARTIFACTS 1.2.0 AND 1.3.0)

Two gates were preregistered on 2026-09-07 in `manuscript/paper15_v12_policy_prereg.md` before any analysis and computed from the frozen 1.1.1 outputs without re-executing any kernel or draw. Amendment A1 records the finding that the 1.1.x column `impact_bal_acc` is the positive part of the signed conclusion change, so that the material label-path count under the prespecified $|\Delta_R| > 0$

TABLE 13

Cluster structure of the frozen evaluation batches (mean over the 30 cells of each environment, standardized projected features): fraction of near-duplicate rows, and the largest fraction of rows within a window of 0.01 or 0.001 standard deviations in any feature (max) or on average over features (mean).

Environment	Dup. rows	Max 0.01	Mean 0.01	Max 0.001
E1 CICIDS ID 256	0.074	0.825	0.562	0.648
E2 UNSW-NB15 ID	0.114	0.735	0.458	0.514
E3 ToN-IoT ID	0.257	0.870	0.466	0.571
E4 Tue→Wed	0.054	0.766	0.489	0.480
E5 Tue→Fri-PortScan	0.129	0.872	0.747	0.822
E6 Wed→Thu-Web	0.004	0.827	0.433	0.696
E7 Wed→Fri-AM	0.026	0.769	0.670	0.663
E8 UNSW temporal OOD	0.100	0.694	0.437	0.477

definition is 2,617 (2,184 lowered, 433 raised) rather than the 2,184 lowered rows alone. The prespecified methodological correction (`manuscript/paper15_v13_prereg.md`) records that the family rule executed in 1.2.0 was asymmetric and that its stated level was false (Proposition 5(c)), adopts the conformal rule of Proposition 5(b), fixes the expectations before regeneration, and discloses that a diagnostic computation of the conformal rule had been seen during the review

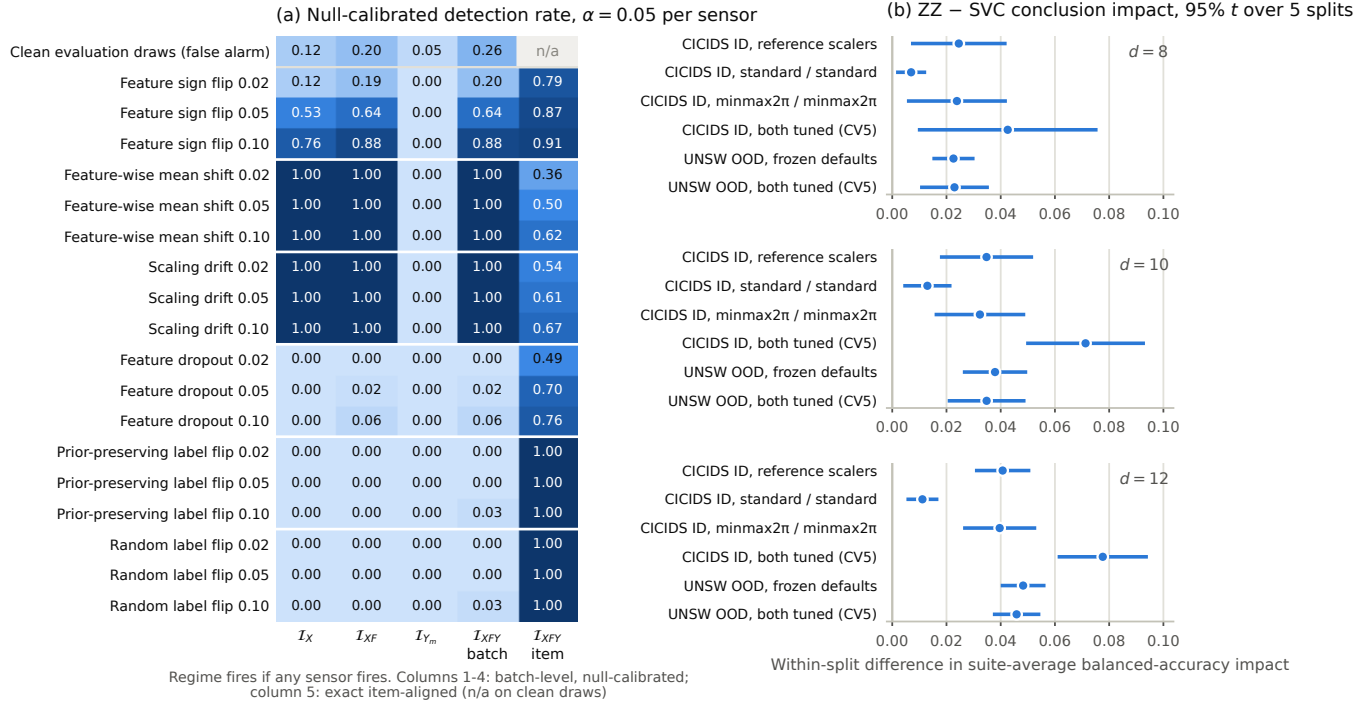


Fig. 2. Prespecified frozen gates. (a) Descriptive finite-batch detection rate of each information regime on the frozen interventions with the empirical false-alarm rate of disjoint clean evaluation draws in the first row; the first four columns are batch-level unions of per-sensor calibrated rules, the last column is exact item-aligned detection (undefined on clean draws). (b) Secondary within-split ZZ-minus-SVC conclusion-impact difference with 95% t intervals over five split clusters under three scaler configurations and under cross-validated tuning of both learners.

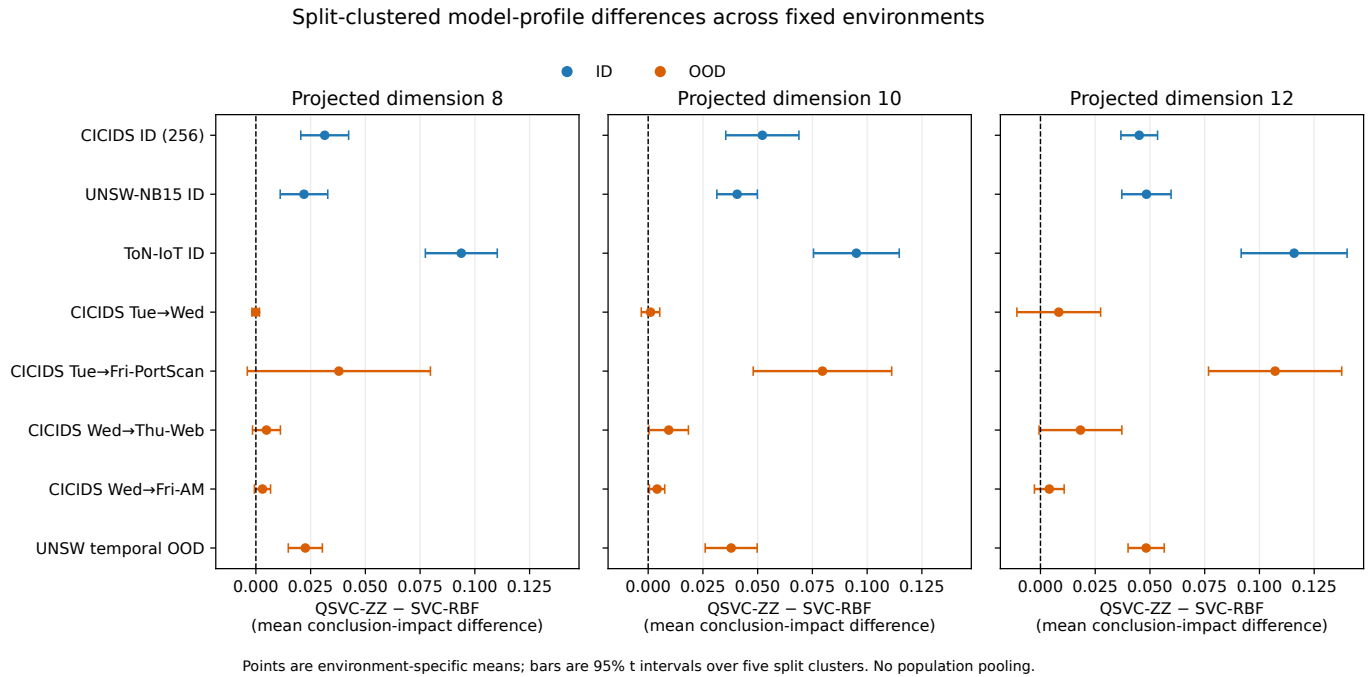


Fig. 3. Environment-specific ZZ-minus-SVC mean conclusion-impact differences and 95% t intervals over five split clusters. ID and OOD settings are fixed cases; no population pooling is performed.

TABLE 14

Gates P and T: within-split ZZ-minus-SVC difference in suite-average balanced-accuracy conclusion impact, mean with 95% t interval over five split clusters and number of positive clusters. CICIDS ID reference is the re-executed P0 configuration; UNSW OOD reference is the frozen 1.0.0 environment E8.

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2 π)	0.0245 [0.0069, 0.0422] (5/5)	0.0348 [0.0176, 0.0520] (5/5)	0.0407 [0.0305, 0.0509] (5/5)
PA standard / standard	0.0070 [0.0014, 0.0125] (5/5)	0.0130 [0.0041, 0.0219] (5/5)	0.0111 [0.0052, 0.0170] (5/5)
PB minmax2 π / minmax2 π	0.0238 [0.0054, 0.0423] (5/5)	0.0323 [0.0156, 0.0491] (5/5)	0.0396 [0.0261, 0.0531] (5/5)
CICIDS ID, both tuned (CV5)	0.0426 [0.0094, 0.0757] (5/5)	0.0713 [0.0494, 0.0932] (5/5)	0.0776 [0.0610, 0.0943] (5/5)
UNSW OOD, frozen defaults	0.0226 [0.0148, 0.0304] (5/5)	0.0379 [0.0260, 0.0498] (5/5)	0.0483 [0.0400, 0.0565] (5/5)
UNSW OOD, both tuned (CV5)	0.0229 [0.0103, 0.0356] (5/5)	0.0348 [0.0205, 0.0492] (5/5)	0.0459 [0.0371, 0.0546] (5/5)

TABLE 15

Gates P and T: clean balanced accuracy (SVC / ZZ, mean over five split clusters) per configuration and dimension, and the distribution of cross-validated regularisation constants over the 30 cells of each tuned environment (SVC also selects γ ; see the artifact table).

Condition	$d = 8$	$d = 10$	$d = 12$
P0 reference (standard / minmax2 π)	0.753 / 0.734	0.747 / 0.728	0.778 / 0.747
PA standard / standard	0.753 / 0.727	0.747 / 0.720	0.778 / 0.727
PB minmax2 π / minmax2 π	0.751 / 0.734	0.750 / 0.728	0.756 / 0.747
CICIDS ID, both tuned (CV5)	0.741 / 0.750	0.746 / 0.781	0.780 / 0.802
UNSW OOD, frozen defaults	0.748 / 0.760	0.754 / 0.766	0.755 / 0.769
UNSW OOD, both tuned (CV5)	0.739 / 0.759	0.723 / 0.757	0.736 / 0.770
Tuned environment	QSVC C selections		SVC C selections
CICIDS ID	$C = 0.1: 3, C = 1: 4, C = 10: 6, C = 100: 17$		$C = 0.1: 1, C = 1: 6, C = 10: 15, C = 100: 8$
UNSW OOD	$C = 1: 17, C = 10: 13$		$C = 0.1: 1, C = 1: 2, C = 10: 15, C = 100: 12$

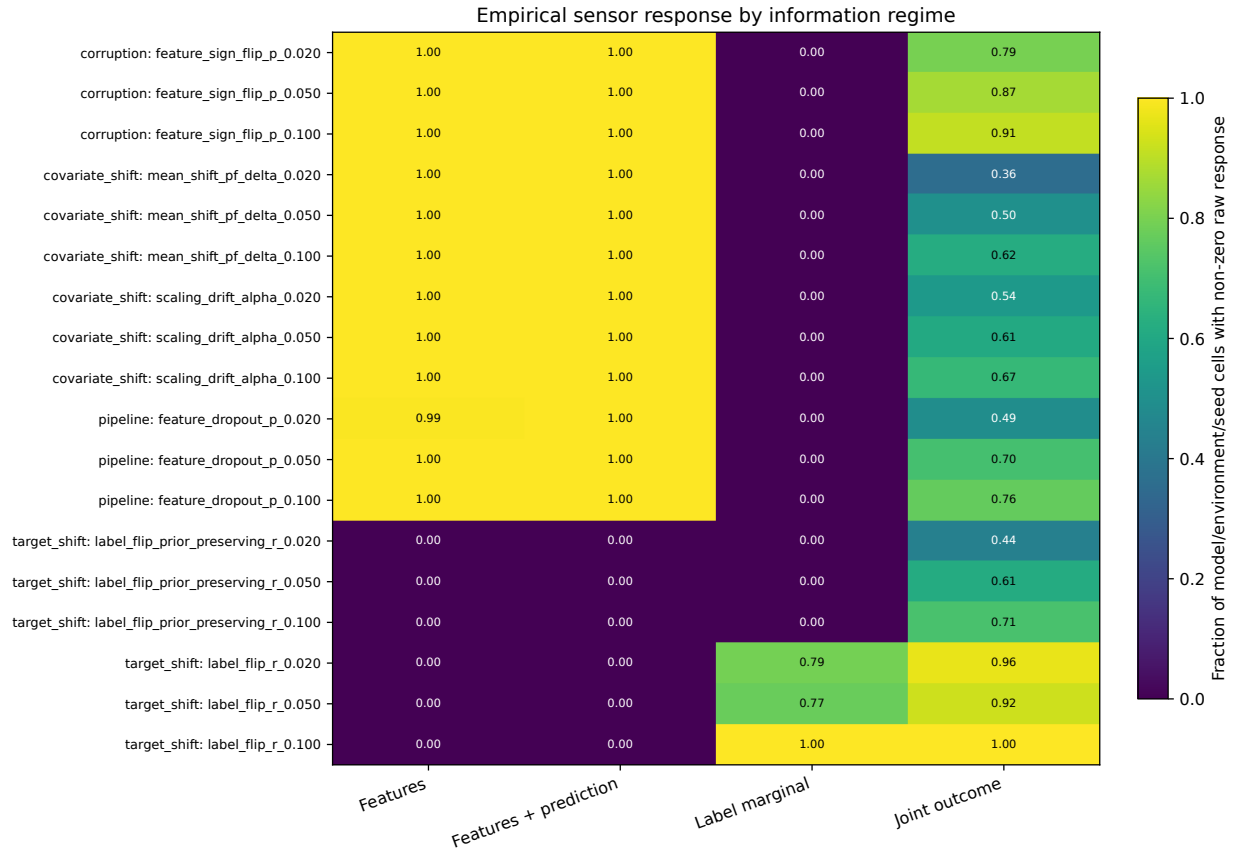


Fig. 4. Raw sensor response by information regime in the frozen expansion. Fractions are over model/environment/seed cells; zero denotes raw invariance, not estimated detector power.

that found the defect (pooled rates 0.056 / 0.058 / 0.048 / 0.053, which the regeneration reproduces exactly). Both gates were regenerated from the same frozen outputs; the 1.2.0 numbers are retained as comparison columns and are used

TABLE 16

Corrected v1.3.7 Gate F on the unchanged null rows. Clean and near-null columns are response rates; the environment range is over the eight clean per-environment means. Re-split uses the same 30 frozen permutations; split construction uses the same fixed draw assignment. All are descriptive for the executed design except the conditional conformal counting statement under exchangeability.

Regime	Clean union	Clean asym.	Clean conf.	Env. conf. range	Near-null conf.	Re-split asym.	Re-split conf.	Split conf.
$\mathcal{I}_X$	0.123	0.063	0.056	0.023–0.113	0.542	0.052	0.045	0.049
$\mathcal{I}_{XF}$	0.201	0.074	0.059	0.024–0.130	0.502	0.058	0.042	0.036
$\mathcal{I}_{Y_m}$	0.048	0.048	0.048	0.025–0.070	0.000	0.040	0.040	0.067
$\mathcal{I}_{XFY}$	0.257	0.080	0.055	0.031–0.135	0.480	0.059	0.035	0.033

for no claim.

11.1 Gate F: Conformal Family Calibration

For every cell and regime the conformal p -value of a batch is computed as in Proposition 5(b) from the 200 calibration draws of the cell, and the regime fires when $p \leq 0.05$. The frozen per-sensor thresholds are reproduced from the calibration draws as a consistency check (F0). Table 16 gives the corrected decision-level response of the union, superseded asymmetric and conformal rules, the range over the eight environments, the near-null controls, the same exchangeable re-splits and the same split construction. The conformal $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules fire on none of the 3,600 label observations and the $\mathcal{I}_{Y_m}$ rule on none of the 1,800 prior-preserving ones (F3); the batch-level $\mathcal{I}_{XFY}$ conformal rule fires on 11 of the 2,617 material label rows (1.2.0 rule 16, union 43). Because these material rows change the confusion view, the 11–43 detections among 2,617 rows measure finite-batch statistical power rather than structural blindness. Structural blind regions are independent of row count, but for interventions that change the observed statistic, power can rise with n as sampling variability often decreases (approximately as $n^{-1/2}$ for proportion-like statistics under regular conditions). Batch-size scaling is an important external-validity question and is not estimated by the present fixed-size experiment. Under the re-splits the conformal rule stays at or below its level in every regime (F4) and the 1.2.0 rule does not.

11.2 Gate D: Offline End-to-End Decisions

Four policies (P0 serve-always, baseline; P1 the uncalibrated union of per-sensor rules, risk-tolerant; P2 the conformal family rule, nominally calibrated under exchangeability and risk-tolerant; P3 the sensor-coverage-complete variant relative to the declared evidence dimensions, which holds whenever a mandatory protected boundary has neither exact nor declared statistical coverage and which guarantees no minimum detection power on the boundaries that are covered) are evaluated over five regimes on 12,000 clean evaluation draws (the 1,200 exact-zero rows for $\mathcal{I}_{XFY}^*$), 1,200 near-null synthetic controls and the 10,800 frozen intervened observations, of which 7,008 are material at $|\Delta_R| > 0$ (1,870 at $\tau = 0.05$). These form 24,000 policy rows derived from the frozen intervention grid; inferential independence resides at the prespecified environment/split clusters rather than at row level. The 12,000 batch clean-decision draws and 1,200 trusted exact-zero invariant rows are different estimands and are never pooled. `block` is reserved for exact violations against a trusted reference; the interruption of the near-null controls is therefore decomposed into statistical holds and

exact-reference blocks. Containment 1.00 follows constructively from Corollary 3 under the declared trusted reference; this experiment measures interruption within the prespecified near-null stress-control grid. In the trusted regime P2 produces 85 gross exact blocks, of which 44 overlap the batch $\mathcal{I}_{XFY}$ /P2 interruptions; its net increment is 41/1,200 (3.42 percentage points), and the trusted total is 617 of 1,200. Table 17 gives the complete decision counts including the hold and block counts on the near-null controls and the non-material interruption rate, Table 18 the conformal-rule policy under the asymmetric comparison and adopted rules, Table 19 the materiality sensitivity and Table 20 the materially altered audit results served by mechanism under P2. The five consistency checks D1–D5 (serve-always allows every material observation; structural blindness is reproduced by the decisions; the trusted regime blocks every material label row and every exact violation; decisions are total and deterministic; the composition reproduces the frozen contract actions) all pass.

12 PREREGISTERED ADVERSARIAL GATE A (FROZEN EXPERIMENT)

Gate A was preregistered in `manuscript/paper15_v13_prereg.md` before execution: threat knowledge, perturbation algorithm, cluster rule (an entry is frozen when at least $\lceil 0.05n \rceil$ rows, never fewer than two, lie within 0.01 batch standard deviations of it; constant features frozen), budget (the unclustered entries, data-determined), strength grid (0.02, 0.05, 0.10 matched to the frozen suite; 0.25, 0.50), matched controls (the executed mechanisms with their frozen tags and seeds), environments, models, seeds, endpoints, acceptance checks, expectations and stopping rule. The 240 jobs completed; the clean rows and matched controls reproduce the frozen expansion exactly (AC1); labels are untouched (AC2); the frozen-entry fraction is recorded on every adaptive row (AC3); every material adaptive row carries a prediction change (AC4). Scoring uses the calibration draws, thresholds, rules and policy layer of Gates F and D. Table 21 gives the corrected-JSD compact overview at matched strengths, while Table 22 reports the unchanged prediction-change and material fractions with the attacker’s budget. The corrected sensor-level, per-strength profiles and the frozen-only without-KS ablation are reported in Tables 28 and 35. The expectation recorded before execution (detection below 0.30 at matched strengths) held for the mean shift in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ at 0.02 and 0.05 and for scaling drift in $\mathcal{I}_X$ at 0.02 and 0.05, and did not hold at 0.10 (0.36–0.65 after the JSD correction); it is reported as executed. No parameter was changed after the results were seen.

TABLE 17

Complete v1.3.7 Gate-D policy counts under the corrected JSD definition. These are deterministic decisions on the frozen rows, not deployed event rates. P3 coverage existence does not imply minimum statistical power.

Regime	Policy	Clean n	False act.	Near-null int.	Material	Allow	Hold	Block	Unsafe allow
$\mathcal{I}_X$	P0	12000	0	0	7008	10800	0	0	7008
$\mathcal{I}_X$	P1 union	12000	1478	904	7008	6356	4444	0	4429
$\mathcal{I}_X$	P2 conf.	12000	677	650	7008	6439	4361	0	4496
$\mathcal{I}_X$	P3 abst.	12000	12000	1200	7008	0	10800	0	0
$\mathcal{I}_{XF}$	P0	12000	0	0	7008	10800	0	0	7008
$\mathcal{I}_{XF}$	P1 union	12000	2417	905	7008	6127	4673	0	4233
$\mathcal{I}_{XF}$	P2 conf.	12000	711	603	7008	6338	4462	0	4368
$\mathcal{I}_{XF}$	P3 abst.	12000	12000	1200	7008	0	10800	0	0
$\mathcal{I}_{Ym}$	P0	12000	0	0	7008	10800	0	0	7008
$\mathcal{I}_{Ym}$	P1 union	12000	570	0	7008	10800	0	0	7008
$\mathcal{I}_{Ym}$	P2 conf.	12000	570	0	7008	10800	0	0	7008
$\mathcal{I}_{Ym}$	P3 abst.	12000	12000	1200	7008	0	10800	0	0
$\mathcal{I}_{XFY}$	P0	12000	0	0	7008	10800	0	0	7008
$\mathcal{I}_{XFY}$	P1 union	12000	3088	905	7008	6076	4724	0	4182
$\mathcal{I}_{XFY}$	P2 conf.	12000	655	576	7008	6381	4419	0	4394
$\mathcal{I}_{XFY}$	P3 abst.	12000	655	576	7008	6381	4419	0	4394
$\mathcal{I}_{XFY}^*$	P0	1200	0	0	7008	10800	0	0	7008
$\mathcal{I}_{XFY}^*$	P1 union	1200	0	922	7008	776	1737	8287	0
$\mathcal{I}_{XFY}^*$	P2 conf.	1200	0	617	7008	857	1656	8287	0
$\mathcal{I}_{XFY}^*$	P3 abst.	1200	0	617	7008	857	1656	8287	0

TABLE 18

Gate D after the JSD correction: P2 under the superseded asymmetric rule and the adopted conformal rule on identical frozen observations. The comparison does not select or replace the primary rule.

Regime	Asymmetric rule (superseded)				Conformal rule (adopted)			
	Clean FPR	Near-null	Served	Contain.	Clean FPR	Near-null	Served	Contain.
$\mathcal{I}_X$	0.063	0.58	4,490	0.36	0.056	0.54	4,496	0.36
$\mathcal{I}_{XF}$	0.074	0.53	4,329	0.38	0.059	0.50	4,368	0.38
$\mathcal{I}_{Ym}$	0.048	0.00	7,008	0.00	0.048	0.00	7,008	0.00
$\mathcal{I}_{XFY}$	0.080	0.52	4,325	0.38	0.055	0.48	4,394	0.37
$\mathcal{I}_{XFY}^*$	0.000	0.55	0	1.00	0.000	0.51	0	1.00

TABLE 19

Gate D after the JSD correction: materially altered audit results served (containment in parentheses) under the three frozen materiality thresholds. The thresholds select reporting endpoints, not attacks. P1 is union and P2 conformal.

Regime	Pol.	$\tau = 0^+$	$\tau = 0.02$	$\tau = 0.05$
$\mathcal{I}_X$	P1	4,429 (0.37)	2,128 (0.44)	652 (0.65)
$\mathcal{I}_X$	P2	4,496 (0.36)	2,174 (0.43)	680 (0.64)
$\mathcal{I}_{XF}$	P1	4,233 (0.40)	1,979 (0.48)	566 (0.70)
$\mathcal{I}_{XF}$	P2	4,368 (0.38)	2,073 (0.45)	628 (0.66)
$\mathcal{I}_{Ym}$	P1	7,008 (0.00)	3,802 (0.00)	1,870 (0.00)
$\mathcal{I}_{Ym}$	P2	7,008 (0.00)	3,802 (0.00)	1,870 (0.00)
$\mathcal{I}_{XFY}$	P1	4,182 (0.40)	1,928 (0.49)	521 (0.72)
$\mathcal{I}_{XFY}$	P2	4,394 (0.37)	2,089 (0.45)	628 (0.66)
$\mathcal{I}_{XFY}^*$	P1	0 (1.00)	0 (1.00)	0 (1.00)
$\mathcal{I}_{XFY}^*$	P2	0 (1.00)	0 (1.00)	0 (1.00)

TABLE 20

Gate D after the JSD correction: materially altered audit results served / material observations by existing intervention mechanism under P2.

Regime	Evaluation-label	Sign flip	Mean shift / scaling	Feature dropout
$\mathcal{I}_X$	2,617/2,617	800/1,464	0/1,848	1,079/1,079
$\mathcal{I}_{XF}$	2,617/2,617	684/1,464	9/1,848	1,058/1,079
$\mathcal{I}_{Ym}$	2,617/2,617	1,464/1,464	1,848/1,848	1,079/1,079
$\mathcal{I}_{XFY}$	2,606/2,617	716/1,464	14/1,848	1,058/1,079
$\mathcal{I}_{XFY}^*$	0/2,617	0/1,464	0/1,848	0/1,079

TABLE 21

Gate A after the JSD correction: the same executed drift controls and cluster-preserving variants, 600 observations per row. Detection is the conformal-family response; materiality and predictions are unchanged. Served is P2 in $\mathcal{I}_{XF}$ among material rows. The construction reduces response against the declared fingerprint; it is not universal evasion.

Mechanism	Strength	Executed mechanism (control)				Cluster-preserving (adaptive, F5)			
		Det. $\mathcal{I}_X$	Det. $\mathcal{I}_{XF}$	Material	Served P2	Det. $\mathcal{I}_X$	Det. $\mathcal{I}_{XF}$	Material	Served P2
Mean shift	0.02	1.00	0.97	0.33	4/197	0.01	0.05	0.29	157/174
Mean shift	0.05	1.00	1.00	0.46	0/275	0.08	0.23	0.40	156/237
Mean shift	0.10	1.00	1.00	0.59	0/355	0.36	0.50	0.54	133/322
Mean shift	0.25	–	–	–	–	0.81	0.83	0.69	52/411
Mean shift	0.50	–	–	–	–	0.88	0.90	0.76	35/454
Scaling drift	0.02	0.99	0.96	0.48	2/290	0.05	0.23	0.40	131/241
Scaling drift	0.05	1.00	0.98	0.57	3/345	0.29	0.42	0.52	104/313
Scaling drift	0.10	1.00	1.00	0.64	0/386	0.57	0.65	0.58	74/350
Scaling drift	0.25	–	–	–	–	0.75	0.80	0.72	53/431
Scaling drift	0.50	–	–	–	–	0.88	0.89	0.81	43/485

TABLE 22

Gate A: prediction-change rate (item-aligned disagreement > 0), material fraction at $|\Delta_R| > 0$, ≥ 0.02 and ≥ 0.05 , mean $|\Delta_R|$, fraction of rows whose balanced accuracy rose, and mean fraction of feature entries the attacker perturbs (the executed mechanisms perturb every entry).

Mechanism	Strength	Pred. change	Mat. 0 ⁺	Mat. 0.02	Mat. 0.05	Mean $ \Delta_R $	Raised	Perturbed
Mean shift (executed)	0.02	0.36	0.33	0.10	0.06	0.011	0.13	–
Mean shift (executed)	0.05	0.50	0.46	0.20	0.11	0.026	0.20	–
Mean shift (executed)	0.10	0.62	0.59	0.35	0.24	0.054	0.20	–
Mean shift (F5)	0.02	0.32	0.29	0.03	0.00	0.003	0.14	0.33
Mean shift (F5)	0.05	0.45	0.40	0.10	0.01	0.006	0.20	0.33
Mean shift (F5)	0.10	0.60	0.54	0.22	0.05	0.014	0.22	0.33
Mean shift (F5)	0.25	0.74	0.69	0.42	0.22	0.033	0.25	0.33
Mean shift (F5)	0.50	0.79	0.76	0.52	0.35	0.053	0.18	0.33
Scaling drift (executed)	0.02	0.54	0.48	0.23	0.14	0.032	0.17	–
Scaling drift (executed)	0.05	0.61	0.57	0.42	0.30	0.063	0.17	–
Scaling drift (executed)	0.10	0.67	0.64	0.50	0.41	0.096	0.19	–
Scaling drift (F5)	0.02	0.47	0.40	0.11	0.02	0.008	0.15	0.33
Scaling drift (F5)	0.05	0.57	0.52	0.27	0.11	0.019	0.16	0.33
Scaling drift (F5)	0.10	0.64	0.58	0.36	0.21	0.030	0.18	0.33
Scaling drift (F5)	0.25	0.77	0.72	0.47	0.33	0.048	0.19	0.33
Scaling drift (F5)	0.50	0.87	0.81	0.53	0.35	0.060	0.18	0.33

13 GEOMETRY-ALIGNED SENSITIVITY

The code audit in `manuscript/METHOD_GEOMETRY_AUDIT_v135.md` found a real but bounded geometry difference. The null-calibrated batch statistics used a fixed frozen evaluation reference E and fresh clean current batches C_k , whereas feature interventions used the same-item comparison E versus $T(E)$. This applies to feature JSD, MMD and KS; score JSD; prediction-rate and prediction JSD; label marginals; and the joint confusion profile. Exact item-aligned prediction disagreement, label mismatch and trusted aggregate deltas were already paired and are not part of this finding.

The follow-up was frozen before results in `manuscript/v135_geometry_aligned_sensitivity_prereg.md`, commit `62cec146689e3ff8cdb95027c7a9c7371ef767fd` at 2026-09-09T14:46:14+02:00. For every frozen model cell it retains $s(E, C_k)$ for calibration and $s(E, B_k)$ for clean-resample evaluation, fixes $B = B_1$, and evaluates every authorized feature intervention as $s(E, T(B))$. Thus reference, batch construction and statistical unit are unchanged between clean and intervention. The 240 deterministic replays cover all eight environments, three dimensions, five split seeds, two model seeds and the original model/map membership: 600 model cells and 13,200 observations over 22 existing identity, near-null, feature-control and adaptive conditions. No label attack, model selection or new random seed is introduced.

Every replay agrees with its frozen clean row within 10^{-9} , all labels are preserved, every expected cell is present and all acceptance checks pass. The exact same-item identity response $s(B, B)$ is zero for all eleven batch scores and four explicit paired scores (maximum 0). The aligned identity copy $s(E, B)$ instead equals the clean-resample row. Its action rate is therefore the clean-resample false-action rate, not an identity false-positive rate: 0.056 in $\mathcal{I}_X$ and 0.059 in $\mathcal{I}_{XF}$.

Tables 29–30 show that mean shift and scaling remain almost completely responsive after alignment. Dropout remains much weaker and strength-dependent. Tiny scaling remains highly responsive; tiny Gaussian is still well above the clean-resample background but is not near-universal. The near-null conclusion is consequently mechanism-specific rather than a single unqualified headline.

The same tables give the prespecified matched Gate A comparison. Across the 12 mechanism/strength/regime comparisons, the adaptive response is lower in 25–40 of 40 environment/split cells, equal in the rest, and higher in none. Adaptive material fractions are 0.262–0.667 at matched strengths. Material nonresponse ranges from 0.077 to 0.822, so materially altered adaptive rows remain uncovered in every matched comparison. At the additional 0.25 and 0.50 strengths, adaptive family response rises to 0.853–0.988 in $\mathcal{I}_X/\mathcal{I}_{XF}$; the result is not strength-independent.

Relative to the original $s(E, T(E))$ design, alignment raises the matched adaptive range from 0.007–0.650 to 0.185–

0.810 while leaving controls at 0.973–1.000. The same-item geometry therefore amplified part of the original contrast. It does not explain away the within-aligned-geometry ordering: the cluster-preserving variant remains consistently below the matched control while retaining material changes. The supported conclusion is correspondingly bounded: calibration controls clean-resample false actions but does not guarantee coverage against this declared monitor-aware attacker. Cluster structure, geometry and adaptive knowledge interact; the experiment does not identify a general causal fraction for any one component. Gate A is a stress test against the declared cluster-sensitive fingerprint, not general evasion of drift detectors; the staged dataset structure contributes to its response, and generalization beyond this fingerprint and structure remains open.

14 VERSION 1.3.7 SENSOR CORRECTION AND LABEL-GEOMETRY CLOSURE

The historical JSD defect is reproducible. Each sample was clipped by its own quantiles, but finite histogram edges were derived only from the reference. When the current sample lay wholly outside those edges, `numpy.histogram(..., density=True)` divided by zero histogram mass and returned NaN; the historical family path then mapped that NaN to $-\infty$. The audit finds 3,374 physical NaNs: 744 direct defective coordinates, 230 derived delta coordinates and 2,400 explicitly non-applicable placeholders. After lineage deduplication these represent 527 scientific evaluations (196 original, 230 aligned-feature and 101 paired diagnostics). Applicable broken values occur in feature JSD under mean shift and scaling drift; the score-JSD helper was vulnerable but has no applicable historical NaN. Every canonical NaN row already had another firing sensor and both family and union fired, so no historical decision is silently reinterpreted. The complete pre-correction audit is `manuscript/JSD_DEFECT_AUDIT_v137.md` and the row ledger is `evidence/jsd_correction/jsd_historical_nan_audit.csv`.

The corrected construction uses reference-derived shared edges, extends the outer bins to $\pm\infty$, forms count PMFs, validates positive total mass and then evaluates JSD. It neither clips a disjoint current sample into the reference range nor uses a special-case fallback. Under valid finite inputs no NaN or infinity reaches calibration; the current path raises on any such value, while the old NaN-to- $-\infty$ rule remains isolated for release reproduction. Across 64,560 corrected primary observation rows, 58,074 scope-qualified row identifiers have at least one changed JSD coordinate. There are 4,432 changed JSD sensor-fire coordinates (4,282 rows), 2,265 changed family-decision coordinates (1,588 rows) and 2,808 changed union-decision coordinates (1,254 rows). The detailed ledger retains old/new sensor values, fires and all four regime decisions.

Table 23 closes the label-side statistical geometry question without changing any structural theorem. The original $s(E, T_y(E))$ geometry reproduces 11/2,617 conformal and 43/2,617 union fires; correcting JSD alone leaves both counts unchanged. Under $s(E, T_y(B))$ with clean comparator $s(E, B)$, the existing interventions produce 2,700 material rows and response rises to 343/2,700 conformal

and 1,183/2,700 union. The denominator changes because materiality is recomputed on the preregistered fresh batch B , not because rows were selected. Among all 2,836 aggregate-separable interventions, the aligned responses are 352 (12.41%) and 1,204 (42.45%). All 764 aggregate-blind rows retain the exact paired-clean sensor vector: their attack-only conformal and union increments are both zero. Thus the label-only blind region, Propositions 1–7, the $R_0/M_0/C$ hierarchy and exact C1 results are unchanged.

The policy consequence in Table 24 is sensitivity-specific, not a retroactive replacement of the original primary grid. For aligned label rows, P2 serves 2,506, 2,509, 2,391 and 2,357 of the 2,700 material rows in $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$ and $\mathcal{I}_{XFY}$, respectively; the trusted regime serves zero. The corresponding union counts served are 2,201, 1,983, 2,391 and 1,517. The corrected-original label counts remain 2,606/2,617 served by $\mathcal{I}_{XFY}/P2$; the all-mechanism primary headline is separately 4,394/7,008 after the JSD correction. Exact item-aligned blocking remains 3,600/3,600 label interventions and 2,617/2,617 original material label rows.

Tables 26–31 expose the sensor decomposition for every frozen mechanism and strength, and Tables 32–36 give the corresponding descriptive ablation. KS rejection is retained in the prespecified primary family. Its dominance is concentrated where the frozen data contain large point masses or tight near-duplicate clusters: small shifts disperse those masses even when other aggregate distances remain modest. The frozen-only without-KS ablation is descriptive and independently reported. In the core grid, 3,521/4,361 primary $\mathcal{I}_X$ family fires, 4,008/4,462 $\mathcal{I}_{XF}$ fires and 3,959/4,419 $\mathcal{I}_{XFY}$ fires survive recalibration without KS; on Gate A the corresponding counts are 5,545/6,401, 6,380/6,835 and 6,251/6,712. By contrast, only 10/650, 62/603 and 62/576 primary near-null family fires survive in those three regimes. Removal can also create different fires because the remaining family is recalibrated; therefore “survives” and the without-KS total are both recorded, and neither replaces the primary.

The confusion-profile L1 presentation in Table 25 is an exact lattice implication of the frozen thresholds, not a power guarantee. At $n = 128$, thresholds 0.203125–0.281250 require 13–19 net confusion-cell count units to exceed them; at $n = 256$, thresholds 0.171875–0.203125 require 23–27. Individual relabelings may cancel in the aggregate, explaining why some $r = 0.02, 0.05$ and 0.10 cells remain below threshold without implying a minimum attack intensity. E1 is the only 256-row environment and has non-zero label response, but batch size is confounded with environment and null behavior, so this contrast cannot identify a batch-size effect.

15 SECONDARY MODEL-IMPACT PROFILE

The ZZ fidelity kernel has the largest mean conclusion impact in the frozen Gate-1 ID suite. Relative to SVC-RBF, mean impact ratios are 2.524, 3.094 and 3.323 at dimensions 8, 10 and 12; paired within-split differences are 0.02427 (95% CI 0.01200 to 0.03654), 0.03289 (0.01998 to 0.04580) and 0.03855 (0.02773 to 0.04937), with all five split clusters positive at every dimension (ZZ/SVC impacts 0.04019/0.01592, 0.04860/0.01571 and 0.05514/0.01659). Across the eight expansion environments the mean ZZ-minus-SVC impact is positive in 7/8 environments at dimension 8 and 8/8 at dimensions 10 and

TABLE 23

Label-side geometry sensitivity. Original uses $s(E, T_y(E))$; aligned uses $s(E, T_y(B))$ with clean comparator $s(E, B)$. Sensor columns are aligned means. Detection is on aligned material rows under $\mathcal{I}_{XFY}$. Aggregate-blind rows preserve every declared label/confusion aggregate.

Attack	r	Mat. orig.	Mat. align.	Sep.	Blind	Label- Δ	Label-JSD	Conf.-L1	Conf.-JSD	Family	Union
prior-preserving	0.02	262	283	283	317	0.038	0.001	0.125	0.005	24/283	118/283
prior-preserving	0.05	369	404	404	196	0.038	0.001	0.131	0.006	46/404	179/404
prior-preserving	0.10	428	421	421	179	0.038	0.001	0.146	0.007	74/421	200/421
random	0.02	521	516	563	37	0.039	0.001	0.127	0.005	54/516	186/516
random	0.05	499	548	599	1	0.040	0.001	0.137	0.006	65/548	244/548
random	0.10	538	528	566	34	0.040	0.001	0.149	0.007	80/528	256/528
Pooled original $\rightarrow$ aligned material response										11/2617 $\rightarrow$ 343/2700	43/2617 $\rightarrow$ 1183/2700

TABLE 24

Label-only P2 sensitivity by geometry. Unsafe allow means a material altered audit result served in this frozen grid, not a deployed event rate. The trusted row uses the unchanged exact item-aligned checks.

Geometry	Regime	n	Material	Allow	Hold	Block	Unsafe allow
original_v136	I_X	3600	2617	3600	0	0	2617
original_v136	I_XF	3600	2617	3600	0	0	2617
original_v136	I_Ym	3600	2617	3600	0	0	2617
original_v136	I_XFY	3600	2617	3589	11	0	2606
original_v136	I_XFY_trusted	3600	2617	0	0	3600	0
corrected_original	I_X	3600	2617	3600	0	0	2617
corrected_original	I_XF	3600	2617	3600	0	0	2617
corrected_original	I_Ym	3600	2617	3600	0	0	2617
corrected_original	I_XFY	3600	2617	3589	11	0	2606
corrected_original	I_XFY_trusted	3600	2617	0	0	3600	0
aligned	I_X	3600	2700	3342	258	0	2506
aligned	I_XF	3600	2700	3342	258	0	2509
aligned	I_Ym	3600	2700	3258	342	0	2391
aligned	I_XFY	3600	2700	3188	412	0	2357
aligned	I_XFY_trusted	3600	2700	0	0	3600	0

TABLE 25

Descriptive lattice implication of the frozen confusion-profile L1 thresholds. Because the statistic lies on a $2/n$ lattice, the table gives the minimum net confusion-cell displacement needed to exceed each cell threshold. Relabelings can cancel in aggregate; these values are not minimum attack intensities or statistical-power guarantees.

Batch	n	Cells	Threshold range	Net count range	Next L1 range
256	6	0.171875–0.203125	23–27	0.179688–0.210938	
128	54	0.203125–0.281250	13–19	0.203125–0.296875	

12; environment-specific intervals exclude zero in 4/8, 7/8 and 5/8; means range from -0.00009 to 0.09382 , 0.00107 to 0.09508 and 0.00403 to 0.11585 (Fig. 3). Clean balanced accuracy and impact correlate positively across expansion cells ($r = 0.714$, exploratory). The preregistered ablations (Fig. 2(b)) show that the direction is stable but the magnitude is conditional: standardizing the quantum branch shrinks the difference to about a third, matched min-max scaling leaves it unchanged, and cross-validated tuning of both learners widens it in CICIDS ID and leaves UNSW temporal OOD unchanged. No pooled population effect is claimed, the profile is neither a quantum-advantage nor a vulnerability ordering, and the conditional-auditability result does not depend on it. Figure 4 shows the raw (uncalibrated) sensor response fractions by regime that motivated the calibration gates.

16 REPRODUCTION AND EVIDENCE MANIFEST

The recommended read-only verification is

```
python -m src.experiments.verify_publication_artifact
--root publication/artifact
```

It verifies the artifact-wide SHA-256 manifest, the eleven embedded evidence manifests and their outputs, row counts, the primary-claim counts including the signed label-path counts, the calibrated label-path consistency checks of the reinforcement gate, the policy-level claims of the regenerated gates (including the exact level of the conformal rule under the exchangeable re-splits), the replay and trust checks of the adversarial gate, and the geometry declaration, identity response, matched pairs and recomputed summaries of the frozen geometry-aligned sensitivity. The complete test command is

```
python -m pytest tests -q -p no:cacheprovider
```

and includes the exhaustive counting-bound tests of Proposition 5(b), the five-vector counterexample to the 1.2.0 rule, the brute-force checks of Lemma 1, Propositions 1–4, 6 and 7 and Corollaries 1–3 on enumerated batches, the intervention-recomputation semantics of the workflow state, the finite-shot counterexample of Proposition 7(iii), and the manuscript consistency gates (abstract length, active headline numbers, conformal and policy wording, the interruption decomposition). The authoritative counts of tests, manifests, outputs, files, pages and hashes for the released version are generated into `publication/RELEASE_STATUS.md` by `src/experiments/make_release_status.py`. The frozen environment uses Python 3.10.16, NumPy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, Qiskit 2.3.0 and Qiskit Machine Learning 0.9.0. Qiskit is pinned below 3 because migrating deprecated feature-map constructors would change circuit serialization and therefore requires a new evidence version.

The artifact omits raw benchmark datasets; it documents their official sources, expected SHA-256 hashes, deterministic staging and a cross-platform hash verifier, and includes source code, locked dependencies, compact derived evidence, figures, figure-source tables, tests, manifests, preregistrations and expected outputs. Every v1.3.5 and v1.3.6 evidence file remains byte-identical to its tagged release. Version 1.3.7 adds only the separate evidence/*jsd_correction* and evidence/*label_geometry_sensitivity* trees under version DOI 10.5281/zenodo.22698329 and concept DOI 10.5281/zenodo.22550852; v1.3.6 remains at version DOI 10.5281/zenodo.22694063. The clean balanced-accuracy table is a presentation-only summary derived from frozen evidence. Source code is licensed under Apache-2.0, derived evidence, figures and documentation under CC BY 4.0, and the manuscript files are author preprints excluded from both licenses.

17 CLAIMS-SUPPORTED / CLAIMS-EXCLUDED MATRIX

Supported by the artifact. Auditability is a property of (intervention class, view, trusted references); structural blind regions are monotone under refinement and closed exactly by added evidence that separates the class; feature/prediction sensors are exactly blind to label-only changes and label marginals to prior-preserving relabeling; every material label-path change alters the confusion matrix and is separable in the joint view, but batch-level Class-A auditors with a benchmark-protected, non-authenticated same-item-set statistical oracle detect 11–43 of 2,617 material rows under the original geometry and 343–1183 of 2700 under the aligned geometry, whereas a trusted aggregate reference of the same batch detects all of them and an item-aligned reference additionally exposes the 808 aggregate-preserving relabelings; the executed rules reproduce the structural blind regions (a consistency check of Proposition 4(i), not a finding); the union of per-sensor rules at $\alpha = 0.05$ yields decision false-action rates of 0.123–0.257 that the conformal family rule brings to 0.048–0.059 in the executed design, whose finite-sample level of 10/201 holds under exchangeability, a

premise the executed design violates by a measured, small margin concentrated in E1; the rule of artifact 1.2.0 had no valid guarantee and most of its published excess was rule bias; on the prespecified equal-weight intervention grid, batch-level regimes with feature evidence serve 4,368–4,496 of 7,008 materially changed results under the conformal-rule risk-tolerant policy, the label-marginal regime serves all of them, and the trusted item-aligned regime serves none while interrupting 617 of 1,200 near-null synthetic controls (85 gross exact-reference blocks, but only 41 net additional interruptions versus batch $\mathcal{I}_{X_{FY}}/P2$); an adaptive attacker who preserves the tight feature clusters cuts detection in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ from 0.94–1.00 to 0.01–0.65 at matched strengths while keeping 83–91% of the conclusion changes, and its materially altered audit results are served by the conformal-rule policy in 27–39% of its material instances in aggregate, with the full profile reported by strength; the geometry-aligned sensitivity retains the matched control/adaptive ordering in all 12 comparisons (25–40 of 40 lower environment/split cells and none higher), while showing that the original same-item construction amplified part of the contrast; the secondary ZZ-minus-SVC profile is direction-stable but scaler- and headroom-dependent; complementary quantum-workflow sensors close different simulator-gate blind regions as instances of one view lattice; and the local prototype executes policy-aware *allow/hold/block* composition offline on frozen outputs.

Explicitly excluded. The evidence does not estimate attack prevalence or population-level deployment incidence, establish universal superiority or vulnerability of quantum kernels, causally isolate feature-map geometry from preprocessing and clean headroom, establish that the exchangeability premise of the calibration holds in deployment, or claim a conditional guarantee given a calibration set. It evaluates one adaptive attacker model and no other. It provides no evidence about QPU security, calibrated device noise, scheduling, provider integrity or attestation, multi-tenancy, side channels, production readiness, deployed runtime services, non-repudiation, persistence, incident handling, recovery under non-stationarity, or operational Fleet Management.

TABLE 26

Original near-null controls: per-sensor decomposition by mechanism and strength. Each calibrated-sensor cell is mean response/firing percentage; KS-stat is descriptive only. Family columns are conformal fires.

Class	Mechanism	Strength	F-JSD	MMD	KS-stat	KS-rej	S-JSD	Δp	P-JSD	Fam. $\mathcal{I}_X$	Fam. $\mathcal{I}_{XF}$
sham	tiny_gaussian	0.001	0.002/0%	0.000/0%	0.174	0.507/65%	0.007/0%	0.000/0%	0.000/0%	275/600	247/600
sham	tiny_scaling	0.001	0.005/2%	0.001/0%	0.297	0.641/85%	0.035/12%	0.001/0%	0.000/0%	375/600	356/600

TABLE 27

Original frozen feature interventions: per-sensor decomposition by mechanism and strength. Each calibrated-sensor cell is mean response/firing percentage; KS-stat is descriptive only. Family columns are conformal fires.

Class	Mechanism	Strength	F-JSD	MMD	KS-stat	KS-rej	S-JSD	Δp	P-JSD	Fam. $\mathcal{I}_X$	Fam. $\mathcal{I}_{XF}$
corruption	sign_flip	–	0.003/0%	0.026/12%	0.020	0.000/0%	0.024/0%	0.035/13%	0.003/13%	54/600	78/600
corruption	sign_flip	–	0.010/0%	0.117/53%	0.050	0.000/0%	0.065/23%	0.083/36%	0.011/38%	281/600	310/600
corruption	sign_flip	–	0.020/0%	0.193/76%	0.093	0.015/0%	0.124/72%	0.133/51%	0.025/51%	430/600	503/600
covariate_shift	mean_shift	0.020	0.065/53%	0.054/18%	0.533	0.959/100%	0.162/63%	0.013/3%	0.001/3%	600/600	584/600
covariate_shift	mean_shift	0.050	0.134/84%	0.122/34%	0.603	0.991/100%	0.253/79%	0.043/11%	0.010/12%	600/600	599/600
covariate_shift	mean_shift	0.100	0.187/90%	0.191/46%	0.623	0.994/100%	0.318/88%	0.089/22%	0.026/24%	600/600	600/600
covariate_shift	scaling_drift	–	0.083/54%	0.087/25%	0.458	0.856/100%	0.225/63%	0.055/15%	0.014/16%	597/600	578/600
covariate_shift	scaling_drift	–	0.152/72%	0.159/37%	0.520	0.890/100%	0.286/78%	0.133/30%	0.048/33%	599/600	587/600
covariate_shift	scaling_drift	–	0.203/91%	0.241/55%	0.564	0.905/100%	0.330/89%	0.186/41%	0.076/43%	600/600	598/600
pipeline	dropout	–	0.001/0%	0.000/0%	0.013	0.000/0%	0.008/0%	0.006/0%	0.000/0%	0/600	0/600
pipeline	dropout	–	0.003/0%	0.000/0%	0.035	0.000/0%	0.023/0%	0.013/2%	0.000/2%	0/600	6/600
pipeline	dropout	–	0.006/0%	0.000/0%	0.062	0.000/0%	0.038/4%	0.024/3%	0.001/4%	0/600	19/600

TABLE 28

Original Gate A matched controls and adaptive variants: per-sensor decomposition by mechanism and strength. Each calibrated-sensor cell is mean response/firing percentage; KS-stat is descriptive only. Family columns are conformal fires.

Class	Mechanism	Strength	F-JSD	MMD	KS-stat	KS-rej	S-JSD	Δp	P-JSD	Fam. $\mathcal{I}_X$	Fam. $\mathcal{I}_{XF}$
adaptive	mean_shift	0.020	0.014/1%	0.000/0%	0.045	0.029/1%	0.043/8%	0.004/0%	0.000/0%	4/600	31/600
adaptive	mean_shift	0.050	0.027/8%	0.004/2%	0.065	0.070/4%	0.068/26%	0.008/0%	0.000/0%	50/600	135/600
adaptive	mean_shift	0.100	0.039/35%	0.019/9%	0.082	0.117/9%	0.090/42%	0.018/3%	0.001/4%	215/600	298/600
adaptive	mean_shift	0.250	0.074/80%	0.067/28%	0.133	0.287/41%	0.131/70%	0.048/16%	0.005/17%	484/600	498/600
adaptive	mean_shift	0.500	0.085/88%	0.144/75%	0.153	0.354/58%	0.153/78%	0.081/34%	0.011/35%	530/600	537/600
adaptive	scaling_drift	0.020	0.021/5%	0.003/0%	0.049	0.032/4%	0.064/26%	0.011/2%	0.001/2%	28/600	135/600
adaptive	scaling_drift	0.050	0.041/32%	0.013/5%	0.069	0.077/8%	0.089/36%	0.029/10%	0.003/10%	173/600	254/600
adaptive	scaling_drift	0.100	0.055/58%	0.036/17%	0.088	0.137/18%	0.106/49%	0.046/17%	0.006/17%	345/600	391/600
adaptive	scaling_drift	0.250	0.074/75%	0.105/56%	0.125	0.260/41%	0.129/61%	0.075/30%	0.012/30%	451/600	479/600
adaptive	scaling_drift	0.500	0.089/86%	0.185/82%	0.154	0.385/61%	0.141/67%	0.098/38%	0.018/38%	525/600	531/600
control	mean_shift	0.020	0.065/53%	0.054/18%	0.533	0.959/100%	0.162/63%	0.013/3%	0.001/3%	600/600	584/600
control	mean_shift	0.050	0.134/84%	0.122/34%	0.603	0.991/100%	0.253/79%	0.043/11%	0.010/12%	600/600	599/600
control	mean_shift	0.100	0.187/90%	0.191/46%	0.623	0.994/100%	0.318/88%	0.089/22%	0.026/24%	600/600	600/600
control	scaling_drift	0.020	0.083/54%	0.087/25%	0.458	0.856/100%	0.225/63%	0.055/15%	0.014/16%	597/600	578/600
control	scaling_drift	0.050	0.152/72%	0.159/37%	0.520	0.890/100%	0.286/78%	0.133/30%	0.048/33%	599/600	587/600
control	scaling_drift	0.100	0.203/91%	0.241/55%	0.564	0.905/100%	0.330/89%	0.186/41%	0.076/43%	600/600	598/600

TABLE 29

Geometry-aligned feature sensitivity: per-sensor decomposition by mechanism and strength. Each calibrated-sensor cell is mean response/firing percentage; KS-stat is descriptive only. Family columns are conformal fires.

Class	Mechanism	Strength	F-JSD	MMD	KS-stat	KS-rej	S-JSD	Δp	P-JSD	Fam. $\mathcal{I}_X$	Fam. $\mathcal{I}_{XF}$
adaptive	mean_shift	0.020	0.042/22%	0.032/11%	0.107	0.073/4%	0.078/27%	0.037/8%	0.002/9%	111/600	169/600
adaptive	mean_shift	0.050	0.048/43%	0.039/14%	0.118	0.135/14%	0.094/45%	0.038/8%	0.002/9%	242/600	327/600
adaptive	mean_shift	0.100	0.056/66%	0.055/28%	0.129	0.195/24%	0.109/62%	0.042/11%	0.003/12%	379/600	437/600
adaptive	mean_shift	0.250	0.077/94%	0.096/50%	0.163	0.364/60%	0.139/80%	0.062/23%	0.006/26%	548/600	575/600
adaptive	mean_shift	0.500	0.090/99%	0.160/80%	0.181	0.406/69%	0.159/86%	0.097/41%	0.013/42%	590/600	593/600
adaptive	scaling_drift	0.020	0.046/34%	0.037/14%	0.111	0.097/10%	0.092/40%	0.039/10%	0.002/10%	184/600	285/600
adaptive	scaling_drift	0.050	0.056/61%	0.052/24%	0.121	0.144/18%	0.108/55%	0.052/18%	0.004/20%	342/600	412/600
adaptive	scaling_drift	0.100	0.065/76%	0.075/39%	0.132	0.206/32%	0.120/64%	0.065/24%	0.007/25%	461/600	486/600
adaptive	scaling_drift	0.250	0.078/85%	0.132/70%	0.157	0.327/55%	0.137/72%	0.091/36%	0.013/36%	512/600	538/600
adaptive	scaling_drift	0.500	0.091/91%	0.197/86%	0.179	0.433/69%	0.148/79%	0.111/43%	0.019/43%	550/600	559/600
control	feature_dropout	0.020	0.039/7%	0.031/11%	0.102	0.042/3%	0.065/8%	0.038/8%	0.002/10%	41/600	38/600
control	feature_dropout	0.050	0.039/9%	0.034/12%	0.111	0.087/8%	0.071/16%	0.042/11%	0.003/12%	52/600	84/600
control	feature_dropout	0.100	0.040/10%	0.041/16%	0.128	0.182/23%	0.076/27%	0.048/17%	0.003/18%	131/600	182/600
control	mean_shift	0.020	0.095/82%	0.083/31%	0.532	0.975/100%	0.188/77%	0.043/9%	0.003/10%	600/600	594/600
control	mean_shift	0.050	0.157/93%	0.149/47%	0.603	0.992/100%	0.269/86%	0.070/18%	0.012/20%	600/600	600/600
control	mean_shift	0.100	0.205/97%	0.217/62%	0.624	0.990/100%	0.329/93%	0.107/30%	0.028/32%	600/600	600/600

TABLE 30

Geometry-aligned feature sensitivity: per-sensor decomposition by mechanism and strength. Each calibrated-sensor cell is mean response/firing percentage; KS-stat is descriptive only. Family columns are conformal fires.

Class	Mechanism	Strength	F-JSD	MMD	KS-stat	KS-rej	S-JSD	Δp	P-JSD	Fam. $\mathcal{I}_X$	Fam. $\mathcal{I}_{XF}$
control	scaling_drift	0.020	0.108/71%	0.114/36%	0.461	0.857/100%	0.244/72%	0.080/20%	0.016/22%	600/600	584/600
control	scaling_drift	0.050	0.167/90%	0.183/51%	0.521	0.895/100%	0.296/87%	0.152/36%	0.050/39%	600/600	595/600
control	scaling_drift	0.100	0.212/95%	0.260/68%	0.564	0.916/100%	0.336/95%	0.200/44%	0.077/47%	600/600	600/600
identity	identity	0.000	0.039/7%	0.031/11%	0.100	0.038/2%	0.063/6%	0.036/7%	0.002/8%	43/600	43/600
near_null	tiny_gaussian	0.001	0.040/13%	0.031/11%	0.197	0.589/81%	0.066/10%	0.036/7%	0.002/8%	366/600	324/600
near_null	tiny_scaling	0.001	0.042/17%	0.032/12%	0.309	0.681/90%	0.088/29%	0.037/7%	0.002/8%	434/600	430/600

TABLE 31

Gate-1 frozen design (response only; no family calibration); per-sensor decomposition by mechanism and strength. Each calibrated-sensor cell is mean response/firing percentage; KS-stat is descriptive only. Family columns are conformal fires.

Class	Mechanism	Strength	F-JSD	MMD	KS-stat	KS-rej	S-JSD	Δp	P-JSD	Fam. $\mathcal{I}_X$	Fam. $\mathcal{I}_{XF}$
clean	unknown	–	0.000	0.000	0.000	0.000	0.000	0.000	0.000	–	–
corruption	sign_flip	–	0.003	0.020	0.018	0.000	0.027	0.045	0.002	–	–
corruption	sign_flip	–	0.009	0.107	0.046	0.000	0.073	0.103	0.009	–	–
corruption	sign_flip	–	0.022	0.192	0.094	0.010	0.142	0.171	0.023	–	–
covariate_shift	mean_shift	0.020	0.058	0.000	0.558	1.000	0.157	0.003	0.000	–	–
covariate_shift	mean_shift	0.050	0.127	0.001	0.604	1.000	0.234	0.005	0.000	–	–
covariate_shift	mean_shift	0.100	0.176	0.015	0.622	1.000	0.323	0.040	0.013	–	–
covariate_shift	scaling_drift	–	0.098	0.001	0.486	0.849	0.323	0.018	0.003	–	–
covariate_shift	scaling_drift	–	0.201	0.032	0.535	0.895	0.383	0.115	0.042	–	–
covariate_shift	scaling_drift	–	0.267	0.116	0.578	0.923	0.421	0.304	0.117	–	–
pipeline	dropout	–	0.001	0.000	0.014	0.000	0.009	0.003	0.000	–	–
pipeline	dropout	–	0.002	0.000	0.034	0.000	0.025	0.006	0.000	–	–
pipeline	dropout	–	0.004	0.000	0.060	0.000	0.041	0.014	0.000	–	–

TABLE 32

Frozen-only descriptive ablation without the prespecified KS-reject component, Original frozen feature interventions. Cells give primary $\rightarrow$ without-KS conformal fire counts; this does not replace the primary family.

Class	Mechanism	Strength	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$
corruption	sign_flip	–	600	54 $\rightarrow$ 54	78 $\rightarrow$ 78	75 $\rightarrow$ 75
corruption	sign_flip	–	600	281 $\rightarrow$ 295	310 $\rightarrow$ 315	283 $\rightarrow$ 284
corruption	sign_flip	–	600	430 $\rightarrow$ 435	503 $\rightarrow$ 505	495 $\rightarrow$ 496
covariate_shift	mean_shift	0.020	600	600 $\rightarrow$ 342	584 $\rightarrow$ 450	583 $\rightarrow$ 441
covariate_shift	mean_shift	0.050	600	600 $\rightarrow$ 518	599 $\rightarrow$ 573	598 $\rightarrow$ 571
covariate_shift	mean_shift	0.100	600	600 $\rightarrow$ 542	600 $\rightarrow$ 590	600 $\rightarrow$ 588
covariate_shift	scaling_drift	–	600	597 $\rightarrow$ 339	578 $\rightarrow$ 398	567 $\rightarrow$ 396
covariate_shift	scaling_drift	–	600	599 $\rightarrow$ 452	587 $\rightarrow$ 500	585 $\rightarrow$ 497
covariate_shift	scaling_drift	–	600	600 $\rightarrow$ 563	598 $\rightarrow$ 581	597 $\rightarrow$ 577
pipeline	dropout	–	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
pipeline	dropout	–	600	0 $\rightarrow$ 0	6 $\rightarrow$ 7	7 $\rightarrow$ 7
pipeline	dropout	–	600	0 $\rightarrow$ 0	19 $\rightarrow$ 20	18 $\rightarrow$ 18
target_shift	label_flip	–	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
target_shift	label_flip	–	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
target_shift	label_flip	–	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	4 $\rightarrow$ 4
target_shift	label_flip_prior_preserving	0.020	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
target_shift	label_flip_prior_preserving	0.050	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	0 $\rightarrow$ 0
target_shift	label_flip_prior_preserving	0.100	600	0 $\rightarrow$ 0	0 $\rightarrow$ 0	7 $\rightarrow$ 7

TABLE 33

Frozen-only descriptive ablation without the prespecified KS-reject component, Geometry-aligned feature sensitivity. Cells give primary $\rightarrow$ without-KS conformal fire counts; this does not replace the primary family.

Class	Mechanism	Strength	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$
adaptive	mean_shift	0.020	600	111 $\rightarrow$ 115	169 $\rightarrow$ 171	167 $\rightarrow$ 171
adaptive	mean_shift	0.050	600	242 $\rightarrow$ 245	327 $\rightarrow$ 325	312 $\rightarrow$ 320
adaptive	mean_shift	0.100	600	379 $\rightarrow$ 384	437 $\rightarrow$ 437	417 $\rightarrow$ 428
adaptive	mean_shift	0.250	600	548 $\rightarrow$ 556	575 $\rightarrow$ 577	570 $\rightarrow$ 571
adaptive	mean_shift	0.500	600	590 $\rightarrow$ 591	593 $\rightarrow$ 592	586 $\rightarrow$ 590
adaptive	scaling_drift	0.020	600	184 $\rightarrow$ 188	285 $\rightarrow$ 285	265 $\rightarrow$ 271
adaptive	scaling_drift	0.050	600	342 $\rightarrow$ 349	412 $\rightarrow$ 416	404 $\rightarrow$ 408
adaptive	scaling_drift	0.100	600	461 $\rightarrow$ 460	486 $\rightarrow$ 485	480 $\rightarrow$ 481
adaptive	scaling_drift	0.250	600	512 $\rightarrow$ 517	538 $\rightarrow$ 539	534 $\rightarrow$ 535
adaptive	scaling_drift	0.500	600	550 $\rightarrow$ 556	559 $\rightarrow$ 561	551 $\rightarrow$ 558
control	feature_dropout	0.020	600	41 $\rightarrow$ 44	38 $\rightarrow$ 41	42 $\rightarrow$ 43
control	feature_dropout	0.050	600	52 $\rightarrow$ 56	84 $\rightarrow$ 83	80 $\rightarrow$ 85
control	feature_dropout	0.100	600	131 $\rightarrow$ 94	182 $\rightarrow$ 160	156 $\rightarrow$ 149
control	mean_shift	0.020	600	600 $\rightarrow$ 506	594 $\rightarrow$ 550	592 $\rightarrow$ 545
control	mean_shift	0.050	600	600 $\rightarrow$ 573	600 $\rightarrow$ 592	600 $\rightarrow$ 592
control	mean_shift	0.100	600	600 $\rightarrow$ 588	600 $\rightarrow$ 598	600 $\rightarrow$ 598
control	scaling_drift	0.020	600	600 $\rightarrow$ 444	584 $\rightarrow$ 485	576 $\rightarrow$ 477
control	scaling_drift	0.050	600	600 $\rightarrow$ 564	595 $\rightarrow$ 579	592 $\rightarrow$ 574

TABLE 34

Frozen-only descriptive ablation without the prespecified KS-reject component, Geometry-aligned feature sensitivity. Cells give primary $\rightarrow$ without-KS conformal fire counts; this does not replace the primary family.

Class	Mechanism	Strength	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$
control	scaling_drift	0.100	600	600 $\rightarrow$ 590	600 $\rightarrow$ 597	600 $\rightarrow$ 596
identity	identity	0.000	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	48 $\rightarrow$ 50
near_null	tiny_gaussian	0.001	600	366 $\rightarrow$ 67	324 $\rightarrow$ 70	313 $\rightarrow$ 70
near_null	tiny_scaling	0.001	600	434 $\rightarrow$ 109	430 $\rightarrow$ 175	414 $\rightarrow$ 177

TABLE 35

Frozen-only descriptive ablation without the prespecified KS-reject component, Original Gate A matched controls and adaptive variants. Cells give primary $\rightarrow$ without-KS conformal fire counts; this does not replace the primary family.

Class	Mechanism	Strength	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$
adaptive	mean_shift	0.020	600	4 $\rightarrow$ 4	31 $\rightarrow$ 31	27 $\rightarrow$ 27
adaptive	mean_shift	0.050	600	50 $\rightarrow$ 49	135 $\rightarrow$ 138	127 $\rightarrow$ 126
adaptive	mean_shift	0.100	600	215 $\rightarrow$ 219	298 $\rightarrow$ 301	280 $\rightarrow$ 282
adaptive	mean_shift	0.250	600	484 $\rightarrow$ 486	498 $\rightarrow$ 501	486 $\rightarrow$ 493
adaptive	mean_shift	0.500	600	530 $\rightarrow$ 530	537 $\rightarrow$ 538	530 $\rightarrow$ 536
adaptive	scaling_drift	0.020	600	28 $\rightarrow$ 26	135 $\rightarrow$ 136	129 $\rightarrow$ 129
adaptive	scaling_drift	0.050	600	173 $\rightarrow$ 180	254 $\rightarrow$ 254	237 $\rightarrow$ 242
adaptive	scaling_drift	0.100	600	345 $\rightarrow$ 337	391 $\rightarrow$ 391	378 $\rightarrow$ 383
adaptive	scaling_drift	0.250	600	451 $\rightarrow$ 453	479 $\rightarrow$ 480	467 $\rightarrow$ 472
adaptive	scaling_drift	0.500	600	525 $\rightarrow$ 529	531 $\rightarrow$ 535	521 $\rightarrow$ 524
control	mean_shift	0.020	600	600 $\rightarrow$ 342	584 $\rightarrow$ 450	583 $\rightarrow$ 441
control	mean_shift	0.050	600	600 $\rightarrow$ 518	599 $\rightarrow$ 573	598 $\rightarrow$ 571
control	mean_shift	0.100	600	600 $\rightarrow$ 542	600 $\rightarrow$ 590	600 $\rightarrow$ 588
control	scaling_drift	0.020	600	597 $\rightarrow$ 339	578 $\rightarrow$ 398	567 $\rightarrow$ 396
control	scaling_drift	0.050	600	599 $\rightarrow$ 452	587 $\rightarrow$ 500	585 $\rightarrow$ 497
control	scaling_drift	0.100	600	600 $\rightarrow$ 563	598 $\rightarrow$ 581	597 $\rightarrow$ 577

TABLE 36

Frozen-only descriptive ablation without the prespecified KS-reject component, Geometry-aligned label sensitivity and original near-null controls. Cells give primary $\rightarrow$ without-KS conformal fire counts; this does not replace the primary family.

Class	Mechanism	Strength	n	$\mathcal{I}_X$	$\mathcal{I}_{XF}$	$\mathcal{I}_{XFY}$
label	label_flip	0.020	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	64 $\rightarrow$ 66
label	label_flip	0.050	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	69 $\rightarrow$ 71
label	label_flip	0.100	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	84 $\rightarrow$ 86
label	label_flip_prior_preserving	0.020	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	48 $\rightarrow$ 50
label	label_flip_prior_preserving	0.050	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	55 $\rightarrow$ 57
label	label_flip_prior_preserving	0.100	600	43 $\rightarrow$ 47	43 $\rightarrow$ 44	92 $\rightarrow$ 95
sham	tiny_gaussian	0.001	600	275 $\rightarrow$ 0	247 $\rightarrow$ 1	243 $\rightarrow$ 1
sham	tiny_scaling	0.001	600	375 $\rightarrow$ 10	356 $\rightarrow$ 63	333 $\rightarrow$ 61